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MIS Quarterly

July 2024 Construct Identity Fallacy Jingle jangle

METHODS ARTICLE

A TOOL FOR ADDRESSING CONSTRUCT IDENTITY IN LITERATURE REVIEWS AND META-ANALYSES[1]

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The problem of detecting whether two behavioral constructs reference the same real-world phenomenon has existed for over 100 years. Discordant naming of constructs is here termed the construct identity fallacy (CIF). We designed and evaluated the construct identity detector (CID), the first tool with large-scale construct identity detection properties and the first tool that does not require respondent data. Through the adaptation and combination of different natural language processing (NLP) algorithms, six designs were created and evaluated against human expert decisions. All six designs were found capable of detecting construct identity, and a design combining two existing algorithms significantly outperformed the other approaches. A set of follow-up studies suggests the tool is valuable as a supplement to expert efforts in literature review and meta-analysis. Beyond design science contributions, this article has important implications related to the taxonomic structure of social and behavioral science constructs, for the jingle and jangle fallacy, the core of the Information Systems nomological network, and the inaccessibility of social and behavioral science knowledge. In sum, CID represents an important, albeit tentative, step toward discipline-wide identification of construct identities.

Keywords: Construct identity fallacy (CIF), construct identity detector (CID), construct validity, nomological networks, synonymy and polysemy, inter-nomological network, jingle and jangle fallacy, natural language processing (NLP), latent semantic analysis (LSA), design science, ontologies

> *The first step of science is to know one thing from another. This knowledge consists in their specific distinctions; but in order that it may be fixed and permanent, distinct names must be given to different things, and those names must be recorded and remembered.*
> Carolus Linnaeus, *Systema Naturae* (1738)

Introduction

The large-scale detection of construct identity has remained a challenging puzzle in the behavioral and social sciences for over 100 years, with *identically* named constructs sometimes representing different real-world phenomena (hereafter referred to as phenomena) and *differently* named constructs often representing the same phenomenon (Kelley 1927; Thorndike 1904). Construct identity fallacy (CIF) is the term coined here to signify discordant naming practices. Suggesting that such puzzles may be the norm rather than the exception, Venkatesh et al. (2003) united eight major theories of technology acceptance after determining that the construct *perceived usefulness* appeared under five different names.

[1]Andrew Burton-Jones was the accepting senior editor for this paper. Yulin Fang served as the associate editor.

The appendices for this paper are located in the "Online Supplements" section of the *MIS Quarterly*'s website (http://www.misq.org).

Addressing unrecognized construct overlaps is "essential for true scientific progress" (Noar and Zimmerman 2005, p. 275) and holds the potential to spark new ideas and connections crucial for the development and integration of theories in social sciences research (Cozby 1993).

This article introduces natural language processing (NLP) for large-scale detection of construct identity and eventual input into finer-meshed factor analyses, and it addresses the research question: How do NLP designs compare to human experts in detecting whether or not two constructs signify the same phenomenon? We developed a tool for use of latent semantic analysis (LSA)—a mathematically based theory of meaning (Deerwester et al. 1990; Landauer 2007)—for construct identity detection, and then tested whether LSA can detect construct correspondence above that of random assignment. We then evaluated LSA against three state-of-the-art designs, which we gave the nomenclature LI, an existing sentence-based similarity algorithm named by our team after its originator (Li et al. 2006); MI, a short-answer grading algorithm, likewise named after its creator (Mihalcea et al. 2006); and Latent Dirichlet Allocation (LDA), a commonly used alternative to LSA (Aggarwal and Zhai 2012).

Tests of the four techniques generated two new approaches, which we termed construct identity detector one and two (CID_{1+2}). CID_1 utilizes the combined principles of LSA and LI. CID_2 utilizes the combined principles of LSA and MI. CID_1 outperformed the other five designs on the tasks of overall construct correspondence detection and on the jangle fallacy detection, whereas LSA outperformed the other designs for the jingle fallacy detection. After evaluations, we combined the best designs into a tool, which we termed the construct identity detector (CID). The design science paradigm (Hevner et al. 2004) applies to research that creates and evaluates tools intended to improve performance in a specific research domain. Specifically, we followed the NLP design science process as described by Abbasi et al. (2012), among others, in framing this article as exaptation research: applying known solutions to new problems (Gregor and Hevner 2013). CID provides a baseline for future research on construct identity detection as well as statistical support for the general hypothesis that NLP-based designs have the potential to perform on par with human experts in detecting whether two constructs signify the same phenomenon.

We assessed CID_1's usefulness empirically by conducting an experiment and a follow-up meta-analytic exercise. We found the tool to be competitive with participants' ability to detect constructs, particularly if the participants were not previously knowledgeable of the constructs that were being evaluated. The study provides the first evidence that large-scale construct identity evaluations are possible, suggesting a path forward for addressing the longstanding jingle (Thorndike 1904) and jangle (Kelley 1927) fallacies that arguably impede the progress of the behavioral sciences (Block 1995).

The next section examines the historical need for and justification of construct identity detection. The following three sections address (1) the proposed methods and evaluation criteria for addressing construct identity; (2) the findings; and (3) discussion and conclusion of the study.

The Construct Identity Fallacy

Thorndike (1904) first described the jingle fallacy as the occurrence of two constructs with identical names referencing different real-world phenomena. Two decades later, Kelley (1927) named the companion jangle fallacy, referencing cases when different construct names are is used to refer to the to the same real-world phenomena. In this article, we combine the two fallacies into the *construct identity fallacy* (CIF), wherein a construct pair references the same or different phenomena and are given dissimilar or identical names, respectively. Construct names are crucial when addressing the CIF, and they provide a common criterion used to find articles for theory reviews (Larsen et al. 2014), one key area in which the CIF impinges upon research progress in the social and behavioral sciences.

Researchers have criticized the

> ever-increasing proliferation of labels that are sometimes offered as synonyms, sometimes presented as specific aspects of the subsuming construct, or, more often, simply loosely used to refer to the related constructs without self-conscious attempts at a more precise or consensual usage (Alexander et al. 1991, p. 315).

Complaints arise from almost every well-researched area in the social and behavioral sciences (Larsen et al. 2013), including *motivation* (Marsh et al. 2003; Murphy and Alexander 2000), *emotional intelligence* (Van Rooy et al. 2005), *mental toughness* (Crust and Swann 2010), *disability* (Pollard et al. 2007), *personality* (Block 1995; Peck 2007), *sensation seeking* (Zuckerman 2008), *user satisfaction* (Zmud et al. 1994), and virtually every subarea related to IS success (Larsen 2003). The field of personality psychology as a whole has been urged to "confront its severe, even crippling, terminological problems" (Block 1995, p. 209). Block (1995) evinced the many difficulties in personality assessment that "derive from the hasty, hazy, lazy use of language" and sug-

Table 1. Construct Identity Fallacy Matrix

		Construct Names	
		Same	**Different**
Phenomena	**Correspondent**	Correct outcome: True positive	Jangle fallacy: False negative
	Independent	Jingle fallacy: False positive	Correct outcome: True negative

gested that "psychologists have tended to be sloppy with words" (p. 209).

A summary of the CIF can be seen in Table 1. Identification of a correspondent pair of constructs with the same names, or identification of an independent pair of constructs with different names leads to correct evaluation outcomes (true positive and true negative, respectively). Alternatively, labeling a pair differently when they are, in fact, correspondent leads to a false negative—the jangle fallacy. Labeling a construct pair the same when they are, in fact, independent leads to a false positive—the jingle fallacy. The next two sections examine the jangle fallacy and the jingle fallacy, respectively.

Jangle Fallacy

> If you commit a jangle fallacy your work is akin to the reinvention of the wheel. Either a lack of knowledge or of concern about the work of other scientists or a lack of required data demonstrating divergent validity between new and established measures/constructs is the cause of the fallacy (Wilhelm 2009, p. 146).

The literature is replete with findings of jangle fallacies. For example, Whiteside and Lynam (2001) have suggested Tellegen's (1982) *control* and Zuckerman's (1994) *disinhibition* scales "seem to measure similar constructs despite bearing different labels" (p. 670). Likewise, the field of self-control research is "significantly handicapped" by jangle fallacies, according to a meta-analysis of 282 studies (Duckworth and Schulze 2009, p. 23). Block (1995, p. 210) also weighed in on the jangle fallacy, stating that it abounds with and is exemplified by instances where constructs are

> put forward without recognition of the earlier and alternative, differently named personality constructs to which they are intrinsically linked or which they blend and confound.

These problems exist in the IS discipline as well. *Perceived usefulness* (Lim and Benbasat 2000) and *switching benefits* (Kim and Kankanhalli 2009) are two constructs used among a large set of very similar IS-discipline constructs with a multitude of names. Assessing both constructs via their measurement items (hereafter referred to as items), both constructs focus on evaluation of a new technology in relation to an existing technology, a distinction that is implicit in Lim and Benbasat's (2000) construct definition and explicit in Kim and Kankanhalli's (2009) definition. For a reader not well-versed in the IS-specific adoption literature, the construct names *voluntariness* (Venkatesh and Davis 2000) versus *extrinsic motivation: external perceived locus of causality* (Malhotra et al. 2008) and their definitions may sound quite different. When evaluating these constructs' items, however, we determined that the two research teams operationalized the constructs inversely (voluntary versus mandatory usage). We argue that the above examples are similar enough that researchers using one construct should be aware of the other.

Jingle Fallacy

> The jingle fallacy applies to any case where two constructs with identical names measure different latent constructs. If you commit a jingle fallacy you transfer available evidence to a new measure although in reality what you conceive as being an instance of an established construct somehow diverges (Wilhelm 2009, p. 146).

The renaming of an existing construct (jangle fallacy) may increase the perceived novelty of a *construct*, whereas the reuse of an existing construct name to represent a different phenomenon (jingle fallacy) may increase the perceived novelty of a *relationship,* as results are likely to diverge from past research. Block (1995) found that,

> within the field of personality psychology, the Big Five factors, as they have evolved and become differently understood while remaining similarly

> labeled by different Big Fivers, represent striking instances of the jingle fallacy (p. 209).

While jangle fallacies were more common in our sample of IS constructs (as shown later), the IS discipline has not been immune to the jingle fallacy. For example, *perceived usefulness* has been used to describe both the perceived importance of skill proficiency on job performance (Nelson 1991) and the perceived belief that a system can enhance job performance (Davis 1989).

Consequences of the Construct Identity Fallacy

There are numerous consequences of the CIF. When researchers search for a construct in an online repository of journal articles, the name of that construct can neither be trusted to elicit high recall[2] (a consequence of jangle) nor high precision[3] (a consequence of jingle). An incomplete understanding of the many names for a construct will lead to literature searches that are incomplete, requiring astute care in guarding against constructs with identical names that reference different phenomena. This added work may, under some conditions, contribute to (1) confusion, misunderstandings, and conceptual messes in research (Alexander et al. 1991; Bong 1996); (2) waste of scientific time (Block 1995; Zuckerman 2008); and (3) difficulty in theory unification and knowledge cumulation (Block 1995; Duckworth and Schulze 2009). Taken together, these consequences hold the potential to stunt progress in the social and behavioral sciences.

Definition and Operationalization of Construct Correspondence

We designate a construct, C, to be correspondent to another construct, C', when both refer to the same real-world phenomenon (Borsboom et al. 2009; Loevinger 1957). Operationally, a construct C will be judged as correspondent to another construct C' if the domain experts determine C could also be used to refer to the real-world phenomenon referenced by C'. The basis for determination might include similarity between indicators such as items, definitions, names, citations, units of analysis, and other evidence for the two constructs. For example, the construct named *perceived performance* was defined by Kim et al. (2009) as "the consumer's perception of how the transaction, including product/ service performance, fulfills his or her needs, wants and desires" (p. 243), and it was operationalized through four measurement items such as "using this website improved my performance in shopping" (p. 254). The construct *utilitarian outcomes* was defined by Hseih et al. (2008) as "the extent to which performing the behavior enhances the effectiveness of personal related activity" (p. 102), and it was operationalized through the four items, including "using the Internet TV improves my performance for communication and information search" (p. 122). Both of these constructs were operationalized at the individual level of analysis, and for both constructs Davis (1989) was cited as a construct source. The domain experts decided that the two constructs were correspondent, and, more specifically, that they provide an instance of the jangle fallacy due to the different names.

Past discussions of the jingle and jangle construct fallacies have strongly depended on whether the two construct names in question were identical or different. While the use of construct names is crucial in framing the historical context of this article's contributions, for methodological purposes, we initially dispense with the notion of name as the identifying feature of a phenomenon in the CIF. Our name dispensation aligns with the role names play in other construct identity methods, such as factor analysis and multitrait–multimethod (MTMM) analysis where construct names serve only as item "containers" (Preacher and MacCallum 2003). Thus, to develop methods that address the CIF, construct names are not strictly needed. We return to construct names during the evaluation phase to separately evaluate design performance for jingle and jangle.

Proposed Methods for Construct Identity Detection

We formulate construct identity detection as a problem of finding constructs that share semantically similar measurement items. We evaluate and use two schools of thought related to semantic and lexical similarity—the corpus model and the knowledge-based model—and use these to develop new hybrid designs (see Table 2). Although the pure corpus and knowledge-based approaches have been described extensively in the literature, their application to large-scale construct identity detection is novel. Details on all six designs are available in Appendix A.

On average and when derived from the same academic domain, the designs should assign a higher similarity score to a correspondent pair than they would to an independent pair.

[2]Precision = true positives / (true positives + false positives).

[3]Recall = true positives / (true positives + false negatives).

Table 2. Summary of Approaches to Construct Identity Detection

Approach	Design	Summary of Design	Justification for Inclusion
Corpus Models	Latent Semantic Analysis (LSA)	A text-similarity method based on a low-rank approximation of a set of context-related texts. Paragraphs from construct articles used for training.	A long history of use in the IS field (e.g., Evangelopoulos et al. 2010; Larsen and Monarchi 2004; Siderova et al. 2008). LSA has proven to be both versatile and effective.
	Latent Dirichlet Allocation (LDA)*	A topic model for texts based on extraction of topics that the examined texts hold in common. Paragraphs from construct articles used for training.	Also a dimensionality reduction technique, but LDA has been found to be superior to other dimensionality reduction approaches (Tirunillai and Tellis 2014), and LDA can be embedded in more complex models (Blei et al. 2003).
Knowledge-based Models	Li et al. (LI)	A sentence-similarity method built upon length and depth distances between words in an existing knowledge base where word order in a sentence is taken into account. WordNet used for training.	Li's method is conspicuous among other methods, as it has been reported to work very well on short texts (Li et al 2006). One of the main reasons is that it preserves word-order information in sentences, which is often lost in other methods.
	Mihalcea et al. (MI)	A sentence-similarity method built upon an existing knowledge base where the similarity score is an aggregation of a number of word-word similarity measures. WordNet used for training.	The method combines different word-similarity metrics and incorporates directionality of word similarity. The method gives specific meaning to words with higher weight than generic concept words. Expected to better differentiate abstract concepts.
Hybrid Models	Construct Identity Detector 1 (CID_1)	Integration of LI and LSA designs. Paragraphs from construct articles used for training.	LI relies on a general English literature knowledge base to derive similarity scores. We speculated that CID_1 would boost LI's performance through use of an LSA domain- specific knowledge base to detect construct properties.
	Construct Identity Detector 2 (CID_2)	Integration of MI and LSA designs. Paragraphs from construct articles used for training.	As the majority of construct items are short, the word specificity in items may convey important information. We expected that narrowing down the specificity to a LSA domain-specific knowledge base would enable CID_2 to distinguish the differences between important concepts.

*LDA was tested based on a reviewer suggestion, and as it did not outperform all other designs, no LDA hybrid model was developed and neither was it included in CID. Future versions of CID may be updated to include LDA. We employed online LDA (Hoffman et al. 2010) through Gensim (Řehůřek and Sojka 2010).

Such designs allow for detection of correspondent construct pairs regardless of the names of these constructs. At each similarity score threshold, we assign any construct pair with a similarity score equal to or higher than the threshold as a correspondent-pair prediction by the design. In the following sections, we briefly discuss the background of the four existing designs and then present the two hybrid designs.

The Corpus-Based Models

Corpus-based models, sometimes known as distributional models, are extensions of vector space models (Salton et al. 1975). Unfortunately, short texts, such as items, do not have sufficient word overlap to create appropriate models (Deerwester et al. 1990; Dumais et al. 1988). Therefore, the corpus-based designs and the two hybrid designs were trained on all paragraphs in 193 academic articles that originated the sample of constructs used in this study.

Corpus-Based Models: LSA

LSA has performed almost as well as humans on complex knowledge management and integration tests (Foltz 2007; Graesser et al. 2007; Landauer 2002). Therefore, LSA repre-

sents a reasonable starting point for evaluation of NLP's potential for coarse evaluation of construct correspondence. Sidorova et al. (2008) and Larsen et al. (2008) detail its application in IS. After creation of an appropriate semantic space, each construct item included in the study was projected into the semantic space to generate its mathematical representation. These representations—item vectors with the same dimensionality as the semantic space—were used to compute cosine angles between all item vectors, with higher cosines indicating higher similarity between items.

Corpus-Based Models: LDA

Latent Dirichlet Allocation (LDA), also known as topic modeling, is a generative algorithm that allows posterior distribution of large collections of documents to reveal the latent semantics of each word in the collections. The underlying idea of LDA is that documents are composed of a mixture of topics, which are represented by words with probabilities related to the topics. For example, a *fruit* topic likely has a high probability of generating food-related words such as *banana*, *orange*, and *grape*. Likewise, these words have high probabilities of being related to the topic of *fruit*.

The Knowledge-Based Models

Knowledge-based models compute sentence similarity from component-word similarities that are derived from human-crafted knowledge bases like WordNet (Miller 1995; Poli et al. 2010). Many knowledge-based models also use corpus statistics and probability theory to measure information content and to indicate whether or not a word is common or specific. Word-order information enables differentiation between such concepts as *information systems* and *systems information*. We appropriate two algorithms (Li et al. 2006; Mihalcea et al. 2006) that are heavily cited and have been reported to be effective in predicting text similarity (Bailey and Meurers 2008; Mitchell and Lapata 2008). Both algorithms use WordNet, which contains a substantial set of English literature words; hence, WordNet does not always cover domain-specific vocabulary, such as that of the IS discipline.

While we limit our scope to the use of WordNet, behavioral researchers interested in developing ontologies or extensions and alternatives to WordNet could use tools such as Protege, the open-source ontology editor. Other ontology-related resources are also emerging that may serve as starting points for discipline-wide ontology development in the future. The controlled vocabulary is a central ontology resource, and constitutes an agreed-upon reduced set of words and phrases that can be used to tag information units or to develop a shared understanding of the entities of interest. Controlled vocabularies are designed to avoid correspondence and independence problems in a domain. Relevant controlled vocabularies include the psychology ontology available on the NIH Bioportal, as well as the National Cancer Institute's Thesaurus (NCIt). Parts of the Systematized Nomenclature of Medicine—Clinical Terms (SNOMED-CT), especially the social context section, also qualify, as do the Medical Subject headings (MeSH)'s Psychiatry and Psychology sections, and the Semanticscience Integrated Ontology available at BioPortal. Iivari et al. (2004), as well as Alter (2005, 2012), propose practical approaches to developing similar content in the IS field.

Knowledge-Based Models: LI

We chose the LI knowledge-based model proposed by Li et al. (2006) because it reportedly works well on short texts. One of the main reasons for its efficacy is the preservation of sentence word-order information, which is normally lost with LDA and LSA. The word order is crucial because it represents syntactic information which, when combined with semantic information, conveys the central meaning of sentences. The LI similarity score is derived from the semantic similarity between words, word-order similarity, and the information content of the words.

Knowledge-Based Models: MI

MI (Mihalcea et al. 2006) computes the similarity score through the combination of word similarity and word specificity. In doing so, it borrows similarity metrics from applications such as malapropism detection.[4] MI also uses word-sense disambiguation (distinguishing between different meanings of a word, such as the word *bank* in the context of river versus finance).

The Hybrid Models

We propose that hybrid models combining the IS-specific domain knowledge of LSA and the word-word similarity components of LI and MI (named CID_1 [LI and LSA] and CID_2 [MI and LSA]) outperform the original designs, because WordNet lacks IS-specific domain knowledge. LI and MI both purport to detect embedded semantics in English-

[4]Detection of the unintentional replacement of one word for another similar-sounding but incorrect word—such as "there's no *stigmata* connected with going to see a psychologist."

language sentences. Yet, whereas the original LI sentence-similarity measure is derived from word order and similarity, MI factors in word similarity and word specificity. Furthermore, both designs use quite different techniques to measure word similarity collectively. Benefiting from specific features of the LI and MI designs, CID_1 and CID_2 use domain-specific and semantically specialized knowledge bases derived from LSA (rather than those from WordNet) to capture the semantic contexts more accurately and thus better detect construct correspondence.

Hybrid Models: Construct Identity Detector 1 (CID_1)

CID_1 is based on combining the LI algorithm with LSA. CID_1 computes item-similarity scores by using LSA's cosine of vectors in the semantic space for word-word similarity rather than the semantic distance of the words in the hierarchical structure of WordNet. We retain the ideas of joint word sets, score normalization, and word ordering. The highly domain-specific and semantically specialized nature of behavioral constructs suggests that replacing the WordNet knowledge base with the LSA-derived domain-specific knowledge base in LI will capture the semantic context of constructs more accurately and therefore better detect construct correspondence.

Hybrid Models: Construct Identity Detector 2 (CID_2)

CID_2 is based on combining the MI sentence-similarity algorithm with LSA. CID_2 computes item similarity by finding the most similar words in the items and weighing the words according to word rarity and commonality. Unlike MI, CID_2 does not rely on WordNet to compute word similarity. CID_2 relies instead on the domain-specific corpus of LSA. This hybrid design replaces MI's six different word-word similarity measures with LSA's singular cosine value.

Evaluation and Findings

Following the collection and categorization of a set of constructs to create an evaluative standard, we evaluated the designs as follows:

1. All six designs (LSA, LI, MI, LDA, CID_1, and CID_2) were evaluated against a gold standard at the construct level, using standard evaluative measures, and then tested for design efficacy under jingle and jangle fallacy conditions.
2. The usability of the best design, CID_1, was evaluated in an experimental setting on four tasks against two groups of Ph.D. student evaluators (participants) as they performed a construct retrieval task using two different systems. This evaluation included an evaluation of common versus uncommon constructs.
3. The usefulness of CID_1 for researching common constructs was evaluated by considering the meta-analytic implications of differences in constructs retrieved.
4. The usefulness of CID_1 for researching uncommon constructs was evaluated by considering the conceptual implications of differences in constructs retrieved.

Design Science Process

One goal for this article was to employ a design science process that should not only serve the purposes of this project, but should enable future evaluation of construct technologies. This process should also allow evaluation of other algorithms proposed for inclusion in CID or of related technologies designed to address the CIF. Addressing the CIF in all the behavioral and social disciplines will require a concerted community effort. We employed two steps, which should also allow future research to move forward with expediency.

Step 1. Procuring constructs and items. The first step in evaluation should be to assess designs against an existing test bed. Because no test bed existed at the time of this project, we had to create one. To simplify future research, we made the construct test bed for this article publicly available. This dataset consists of the construct names, definitions, and all measurement items, as well as construct-level classification information. Future designs should show an ability to outperform the designs tested in this article, either on the test bed from this article, a larger or different IS test bed, or on a test bed developed for another discipline. The designs tested would not necessarily need to outperform existing designs for all available test beds since, over time, designs should be optimized for specific settings. For effective progress, it is important for new test beds or reanalysis of existing test beds to be shared as public resources. Because no two researchers will ever completely agree on the structure of a sufficiently complex taxonomy, improving taxonomies themselves should be considered worthy research efforts—as long as the improved taxonomy is shared in full, in machine-readable format.

Step 2. Developing new designs or importing existing designs. Following Gregor and Hevner (2013), design science work addressing the CIF should fall into the exapta-

Table 3. Design Science Contributions

Guideline (Hevner et al. 2004)	Contribution
Design as an Artifact	The research outcomes: 1. CID methodology. 2. The extension and operationalization of jingle and jangle fallacies. 3. The CID tool for integration of theories.
Problem Relevance	Large-scale construct identity problem was identified over 100 years ago with no viable solution to date, in spite of increasing relevance due to publication volume. Providing tools for recognition of overlaps between theories is essential for true scientific progress (Noar and Zimmerman 2005).
Design Evaluation	Utility and efficacy of design outcome evaluated using expert gold standard, experiment with Ph.D. students, a meta-analytic evaluation of common constructs, and a conceptual analysis of uncommon constructs.
Research Contributions	The first design tool, showing promise for detecting construct correspondence and independence. Expanded knowledge about IS constructs.
Research Rigor	Comparison of six designs using a carefully developed gold standard, as well as other evaluations, resulting in triangulation.
Design as a Search Process	Discovery of an effective solution in the form of CID_1. Evaluation of two different types of similarity designs and the creation of hybrid designs.
Communication of Research	Detailed information on how to build replicable CID technologies and implications for behavioral research.

tion or improvement categories. When addressing a new discipline where CID designs have not been used before, exaptation research should suffice for the first article, and research findings would be valuable even if only existing designs are used. In such cases, the contribution will consist of examining the extent to which semantic and lexical designs have the potential to work in the focal discipline, and the designs discussed herein may represent a reasonable first step. For improvement research, the focus will be on ensuring that these designs perform better than existing designs applied to construct identity detection in the focal discipline. Table 3 outlines the design science process and contributions of this article in the context of Hevner et al.'s (2004) criteria.

Gold Standard Evaluation

Evaluating the six designs in this article required a gold standard. A gold standard—"data that is hand-annotated by domain experts" (Cohen and Hunter 2004, p. 153) who share an understanding of the task—is used as the "solution" against which the design performance is assessed and represents a taxonomy's use for machine learning training or evaluation. As is common in design science research (Gregory and Muntermann 2014; Simon 1981), we utilized several heuristics during gold standard creation. First, our experience has shown that occasionally, construct definitions and items imperfectly mirror each other. Given that items undergo rigorous testing and are "closer" to real-world phenomena than are definitions, we asked experts, in discrepant cases, to attend primarily to items. Second, we have found no discernable rules for whether to explicitly include *perceived* as part of a construct name. When supported by the definition and items, experts were asked not to use the term *perceived* in a construct name as a distinguishing feature for determining construct correspondence.

Three final heuristics addressed construct ambiguity, including congeneric methods effects; mono-method bias (Ortiz de Guinea et al. 2013); items signifying more than one concept (Abbott 1997); ambiguity of an item's meaning (Abbott 1997); nonlinear and complex relationships between constructs (Abbott 1997; Ortiz De Guinea et al. 2014); the linguistic nature of questionnaire processing (Arnulf et al. 2014); complex constructs encompassing emotions, cognitions, and behaviors (Ortiz de Guinea and Webster 2013); and use of individual-level items to evaluate group-level constructs (Burton-Jones and Gallivan 2007; Easley et al. 2003). First, because the jingle fallacy cannot conclusively be assessed at the item level, domain experts were asked to rigorously treat overlapping but ambiguous constructs as independent, rather than as correspondent, resulting in a conservative evaluation (Burton-Jones and Gallivan 2007; Easley et al. 2003). Second, while any individual indicator or subset of indicators may be ambiguous, we included only constructs with three or more items reported. Third, we averaged item similarities to aggregate results to the construct level.

Table 4. The 20 Most Common Construct Clusters

Cluster Name	Number of Constructs	Cluster Name	Number of Constructs
Individual-level usefulness	67	IS project success and performance	11
Ease of use	46	Social influence	10
Intention to use	29	Actual user participation	10
Affect towards technology use	28	Technology compatibility with work style/values	10
Facilitating conditions	23	Job overload	9
Individual-level trust: Trustworthiness	19	Information quality	9
General concerns about information privacy	16	Credibility: Trust in ability and benevolence	9
Satisfaction with technology	15	Individual-level trust: Trust propensity	9
Technology self-efficacy	12	Trust in others' ability	8
Organization-level usefulness	12	Satisfaction with decision-making outcome	8

The constructs for this article were collected from two journals, *MIS Quarterly* and *Information Systems Research*, which were selected for their broad nature and the higher-than-average expectation for research novelty in highly ranked journals. The research team considered these journals likely to contain a broad set of constructs consistent with the interests of many researchers in the discipline. After examining articles published during the 1983 to 2009 period—and after selecting articles containing at least one construct that explicitly reported at least three items for inclusion—193 articles were found, with a total of 1,004 constructs.

The 1,004 constructs were categorized into 347 correspondent clusters with an average of 2.89 correspondent constructs in each cluster (S.D. = 5.32). In this dataset, 178 constructs (17.7%) were one-offs that never gained traction in these journals. Correspondent constructs may exist for most of these singlets outside of the two journals examined. The 1,004 constructs yielded 503,506 construct pairs, including 5,851 correspondent construct pairs. Table 4 lists the 20 largest construct clusters, and the entire gold standard is reported in Appendix B.

To evaluate the extent of the construct identity problem in the context of the combinatorics necessary to address the problem, we separately categorized names for the 1,004 constructs as same or different and compared these results to the gold standard categorization of the same constructs. The term "perceived" appeared in many construct names, which indicates that the construct is ontologically subjective, whereas the lack of "perceived" could indicate a case of an ontologically objective construct (Burton-Jones 2009). This gold standard set contained predominantly ontologically subjective constructs. Therefore, we removed the word "perceived" from all the constructs names to avoid falsely inflating the number of unique construct names. Doing so led to removal of 115 instances of "perceived." Table 5 shows the results.

For the understanding of problem significance, the true negatives (497,236 pairs) are misleading. In fact, any classification task with hundreds of categories will predominantly generate true negative pairs by the nature of the combinatorics. After all, researchers generally do not start with a construct such as *usefulness* and evaluate it against all other constructs. Literature reviews are generally based on locating constructs that correspond to the constructs in which one is interested. As displayed in Table 5, researchers looking for correspondent constructs will, when attending to names, return independent constructs 28% of the time. Seldom will researchers find the jangle fallacy constructs, which constituted 82% of the total set of correspondent pairs in the gold standard.

NLP Design Findings

We used each of the six designs to generate similarity scores for the 11,270,591 relationships between 4,751 items, which were then aggregated into construct-level relationships (see Appendix C) and evaluated against the expert-generated gold standard. We report the outcome using precision, recall, and F_1-scores and receiver operating characteristic (ROC) curves, which are plots that show the true positive rate against the false positive rate at multiple cutoff points, for which we conducted statistical significance tests. Finally, we calculated the area under the curve (AUC) for each ROC curve.

The ROC plotting (Figure 1) was based on the information collected at 21 similarity-cutoff thresholds in 0.05 increments

Table 5. Construct Identity Fallacy Occurrences

	Construct Names	
	Same	**Different**
Correspondent	True positive: 1,053 pairs	Jangle fallacy: 4,798 pairs
Independent	Jingle fallacy: 419 pairs	True negative: 497,236 pairs

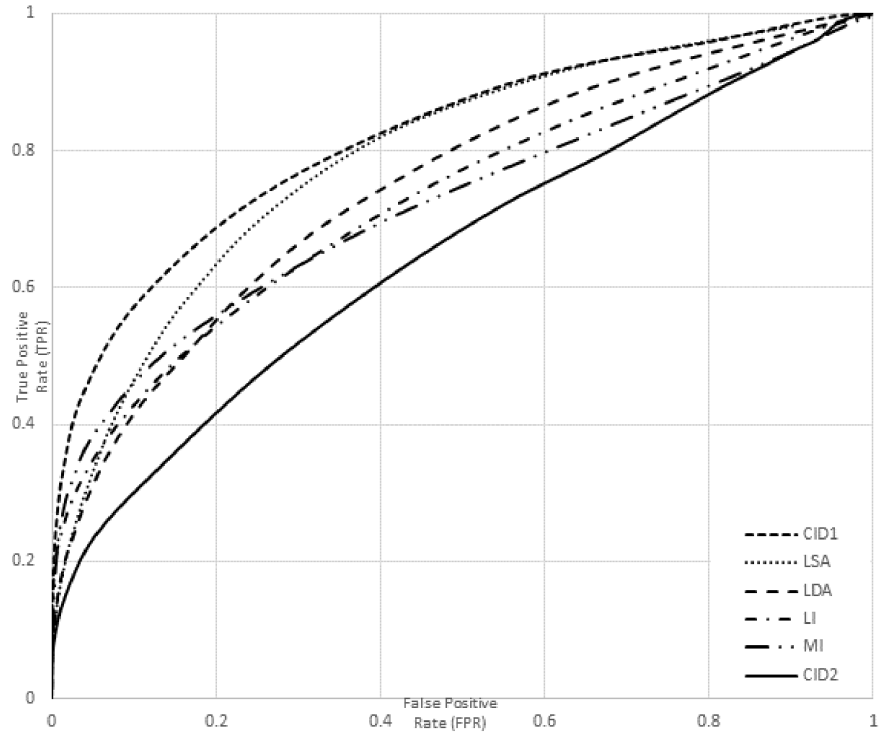


Figure 1. ROC Curves for All Designs

ranging from 0.0 to 1.0. At each cutoff threshold, we consider any construct pair with a similarity score at or above the cutoff to be correspondent, whereas, we consider any pair below the cutoff to be independent. Therefore, we can simply sort the construct pairs according to the similarity score and count the numbers of true positives (TPs) and false positives (FPs), depending on whether the predicted relationships are in agreement with the gold standard.

Comparing the AUCs of the six designs (see Table 6) using the concentrated ROC framework (CROC) (Swamidass et al. 2010) confirms that LSA does indeed perform significantly better than random assignment as well as LI and MI, and that, as expected, CID_1 significantly outperforms LSA, LDA, LI, and MI at $p < 0.01$.

Focusing on the best original design (LSA) along with CID_1 and LDA, Figure 2 summarizes precision and recall evaluation by presenting the cumulative frequency distribution histogram of the correspondent constructs at 21 different cutoffs, with the X-axis showing various intervals of similarity scores (e.g., the interval labeled 0.50 includes any score from 0.500 to 0.549). The few negative LSA cosines were truncated to zero. The rightmost Y-axis of the figure represents the number of correspondent constructs discovered. The precision of correspondent constructs obtained at different cutoff scores is plotted as lines and overlaid on the histogram in the left Y-axis.

Looking at Figure 2, we see that CID_1 has the highest precision at higher cutoffs (as compared to LSA and LDA).

Table 6. Area Under the Curve of ROC and CROC Tests

Design	AUC*
CID_1	**0.814**
LSA	0.791**
LI	0.728**
MI	0.718**
LDA	0.680**
CID_2	0.643**

*CROC significance evaluations were conducted for all algorithms against CID_1.

**$\alpha < 0.01$ on the ratio of true positive rate versus false positive rate compared to best design (CID_1).

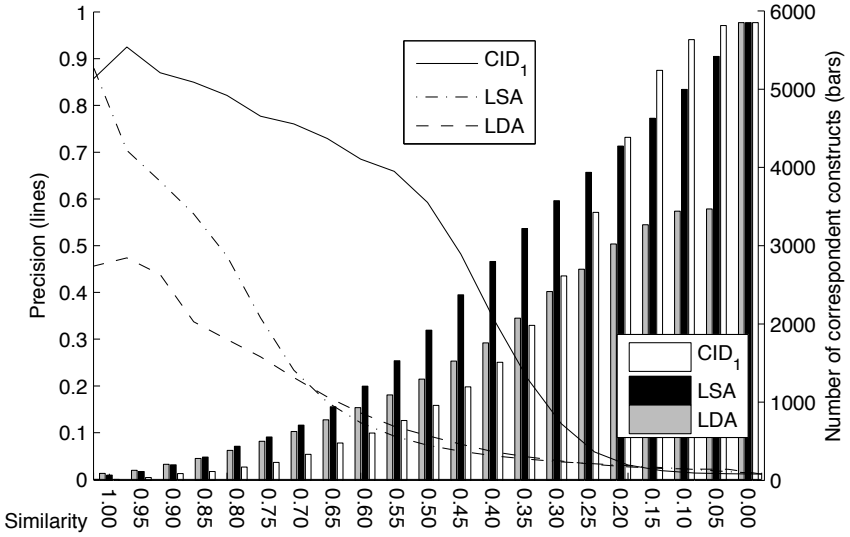


Figure 2. Cumulative Number of Correspondent Constructs Found

Although its precision decreases at a faster rate after the similarity cutoff of 0.50 when compared to LSA, CID_1's precision remains higher than that of the other designs down to a cosine threshold of 0.15. LSA's strong initial performance, compared to LDA on both precision and detected correspondent pairs, leads to a better overall performance and a second rank on AUC (Table 6).

From Figure 3, we can see that LI performed nearly on par with LSA and outperformed MI and CID_2, with CID_2 as the weakest performer. Thus, with respect to finding correspondent constructs, CID_1 is a better design because it has higher precision and results in more correspondent pairs. Significantly, random assignment would yield a precision of 0.012, which, by comparison, CID_1 outperforms by a factor of 48 or higher all the way down to the .50 similarity cutoff. In fact, CID_1 finds almost all correspondent construct pairs before nearing the random assignment precision level.

Both CID_1 and CID_2 frequencies were normally distributed,[5] whereas LSA's distribution is skewed to the right and not normal. Figures 2 and 3 demonstrate that a better design is one that has high precision as well as a high number of correspondent pairs above higher cutoff points. The actual cutoff value is of little interest here, as one can assume that a

[5]Normalities were tested with the Kolmogorov–Smirnov test.

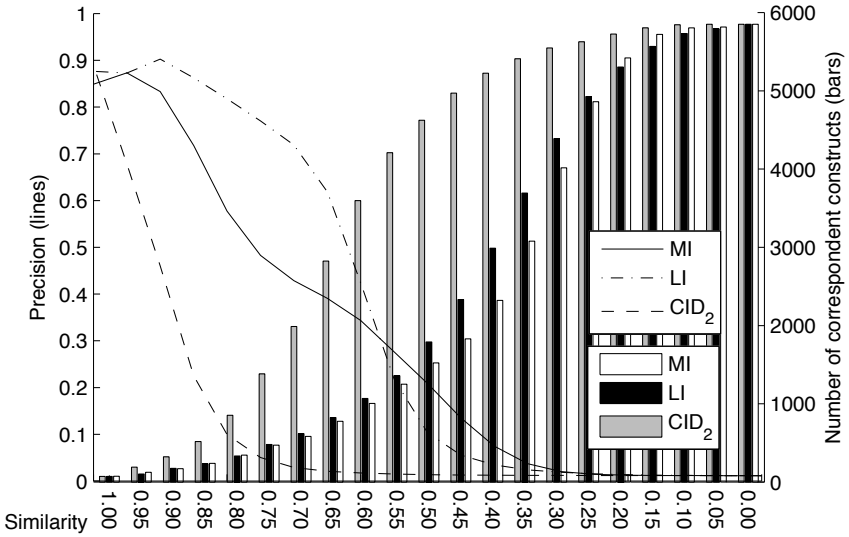


Figure 3. Cumulative Number of Correspondent Constructs Found (Continued)

future user of CID has a fixed level of patience with false positives and will continue to review results until the precision rate gets too low. CID_1's performance is favorable due to its better balance between precision and number of returned correspondent constructs. Despite CID_2's capacity for identifying a high proportion of correspondent pairs at higher cutoff levels than the other designs, its high false positive rate renders it an inferior design.

Jingle and Jangle Fallacy Findings

Moving into the CIF components, a test of the overall performance of the six designs for addressing the jingle and jangle fallacies was conducted in the same way by plotting ROC charts. Doing so presented a problem because, per Table 1, a case of jingle fallacy represents a false positive. The existence of true negatives complicates the detection of these false positives. In this case, true negatives represent more pairs in excess of two orders of magnitude (see Table 5), which would severely reduce the diagnosticity of any test. We therefore removed all true negative pairs and evaluated each design's ability to provide low similarity scores for jingle fallacy pairs (pairs with same names for independent pairs). We inverted similarity scores (1-sim) to simplify presentation of ROC chart and the AUC calculations.

The jangle fallacy provides an equivalent, if smaller, problem in that for every five jangle fallacy pairs (false negatives) there is about one true positive pair. Starting with the original dataset, we therefore removed all true positive pairs and reevaluated all designs for their ability to provide high similarity scores for jangle fallacy pairs. As demonstrated in Table 7, we were able to address the jangle fallacy through the designs we examined. Removing the true positives (constructs with the same names measuring the same phenomenon) would likely retain the construct pairs that are hardest to detect. In spite of this, CID_1 retained the vast majority of its ability to detect correspondent pairs. As may be seen by comparing jangle fallacy AUC scores (Table 7) to overall AUC results (Table 6), the order of performance remained consistent, and CID_1 lost less ability than the other top designs,[6] only 0.0158 AUC. CROC evaluation showed that CID_1's performance was significantly better than all other designs. Further, Table 7 shows that we were able to address the jingle fallacy using CID designs. While AUC performance is lower than that for the jangle fallacy and lower than overall performance, the performance is still respectable. In this case, LSA significantly outperformed all other designs; CID_1 came in second and also significantly outperformed all other designs.

[6]Only CID_2 lost less ability, but this is likely due to the poor initial performance of CID_2.

Table 7. Area Under the Curve (AUC) of ROC and CROC Tests for Jingle and Jangle Fallacies

Jangle Fallacy		Jingle Fallacy	
Design	**AUC**	**Design**	**AUC***
CID_1	0.7982	LSA	0.7328
LSA	0.7735**	CID_1	0.6719**
LI	0.6986**	LI	0.6424**
MI	0.6939**	MI	0.6050**
LDA	0.6565**	LDA	0.5809**
CID_2	0.6380**	CID2	0.5767**

*CROC significance evaluations were for the jingle fallacy conducted for all algorithms against LSA.

**$\alpha < 0.01$ on the ratio of true positive rate versus false positive rate compared to best design (CID_1).

While the above evaluations clearly showed the efficacy of the CID_1 and LSA designs, the practical implications of such designs bear evaluation. Therefore, we conducted a construct retrieval experiment in which CID_1 was evaluated against 36 doctoral students who were randomly assigned to use either a full-text search engine that is commonly available in libraries worldwide (EBSCO), or a newer search engine containing extracted constructs (INN). The experimental evaluation section contains the full evaluation procedure and results.

Experimental Evaluation

To evaluate the usability and usefulness of CID_1, we conducted a randomized experiment with students from a wide range of Ph.D. programs in the United States and around the world. After evaluating design performance by participants on four construct-detection tasks, we evaluated the performance of CID_1 for common constructs and then for uncommon constructs.

Experimental Process

To evaluate CID_1's ability to perform large-scale retrieval of constructs, we selected EBSCO HOST's full-text article search because it is widely used in practice, and equivalent or better performance by CID_1 would suggest practical merit. At the time of the experimental process, EBSCO further represented the longest uninterrupted period of full-text coverage for both *MIS Quarterly* and *Information Systems Research* available to the research team (1996–2009). The Inter-Nomological Networks (INN)—an online Web portal enabling direct search for constructs (Larsen and Bong 2015) in journals such as *MIS Quarterly* and *Information Systems Research*—also covered the same period with a total of 2,908 variables, including 1,730 behavioral constructs that we classified according to the process outlined in the evaluation section. INN provides the opportunity to do a theoretical comparison because it shows what a search platform like EBSCO could achieve if designed more effectively, and it enables comparison of expert judgment versus automatic design results on similar database content, favoring the experts. Expert advantages came from having access to construct names, definitions, and citation information as well as the item information used by CID_1. We built a special-purpose version of INN that enabled keyword search only for the selected period, using the Lucene open-source search engine (Hatcher et al. 2004). Arguably, removing all the misleading full-text around constructs will account for much of the ability of the designs to outperform experts using full-text search.

Using t-tests, the two experimental groups were evaluated based on *age, years of work experience, years in Ph.D. program*, *have/do not have Master's degree, experience with construct-based research*, and *task experience,* none of which was significant at $p < .05$, suggesting no major differences between groups. Two variables failed the Poisson test and were analyzed via ranked scores using a Mann–Whitney test. The self-reported *number of articles published in journals* was not significantly different between the groups, whereas the *number of articles published in conferences* was significantly higher for the group assigned to the EBSCO system ($z = -2.10$, $p = 0.036$), suggesting that any difference between the groups favored the traditional approach to construct retrieval. One additional indicator suggested an advantage for the traditional system: the average participant of experimental group 1 (EBSCO) indicated that s/he had used the system more than six times before, whereas participants in experimental group 2 (INN) had, for the most part, never used the system ($\chi^2 = 15.451$, $df = 4$, p-value = 0.004), suggesting a significant advantage for the traditional system.

Table 8. Constructs for Tasks 1 through 4

Task/Construct*	Definition	Items
1. **Social Influence** (Hsieh et al. 2008) (32 correspondents)	The perceived expectation from family, relatives, friends, and peers for an individual to perform the behavior of interest	• My family thinks that I should use the Internet TV • My relatives think that I should use the Internet TV • My friends think that I should use the Internet TV • People I work with think I should use the Internet TV
2. **Perceived usefulness** (Chin et al. 2003) (71 correspondents)	The degree to which a system is perceived to enhance one's performance	• Using Electronic Mail in my job enables (would enable) me to accomplish tasks more quickly • Using Electronic Mail improves (would improve) my job performance • Using Electronic Mail in my job increases (would increase) my productivity • Using Electronic Mail enhances (would enhance) my effectiveness on the job • Using Electronic Mail makes it (would make it) easier to do my job • I find (would find) Electronic Mail useful in my job
3. **Familiarity [with website]** (Gefen 2000) (4 correspondents)	Consumer's familiarity with the site	• Overall, I am familiar with this site • I am familiar with searching for items on this site • I am familiar with the process of purchasing from this site • I am familiar with buying products from this site
4. **Errors [in personal data held by organization]** (Smith et al. 1996) (3 correspondents)	Consumers' opinions about errors in organizational practices	• All the personal information in computer databases should be double-checked for accuracy—no matter how much this costs • Online companies should take more steps to make sure that the personal information in their files is accurate • Online companies should have better procedures to correct errors in personal information • Online companies should devote more time and effort to verifying the accuracy of the personal information in their databases

*The number of correspondent constructs differs from those in Table 4 due to different inclusion criteria imposed by the period of full-text availability for EBSCO.

We evaluated CID_1's performance against that of the experimental groups using precision, recall, and F_1-measures. CID_1 design answers cannot be considered equivalent to individual human responses; furthermore, the design produces only one answer, meaning that statistical significance tests comparing the design and the groups of human participants were not possible. In spite of this, we believe the results (1) allow evaluation of the usability of the designs for retrieval of constructs and (2) offer suggestions as to circumstances under which a specific design may best approximate human capabilities (keeping in mind the impossibility of human-coder processes for the problem of construct-identity detection, given the sheer volume of published constructs).

Tasks

Because construct-retrieval tasks are very time consuming, four tasks were provided for the participants, two common tasks (commonly used and likely known constructs) and two uncommon tasks (infrequently used and likely unknown constructs). The tasks were randomly selected from the pools of classified constructs, two from the set of common constructs and two from the pools with only three to five synonymous constructs (uncommon constructs) (see Table 8). For each task, the participants were given, as the starting point, an example of a construct name along with a definition and a set of items. Participants were then asked to find all synonymous constructs in the given database. The starting construct example was randomly picked from each pool of synonymous constructs. Participants were randomly assigned to start with common or uncommon tasks; we found no statistical difference in performance based on this order at $p < .05$. On average, participants self-reported that each task took 29.7 minutes, with common tasks at 32.8 minutes and uncommon tasks at 26.6 minutes.

The first author and a chief research assistant with four years of experience in construct extraction and evaluation assessed the first half of all non-categorized constructs, which were those answers by the participants that could not immediately be resolved as correct (primarily an issue with the EBSCO

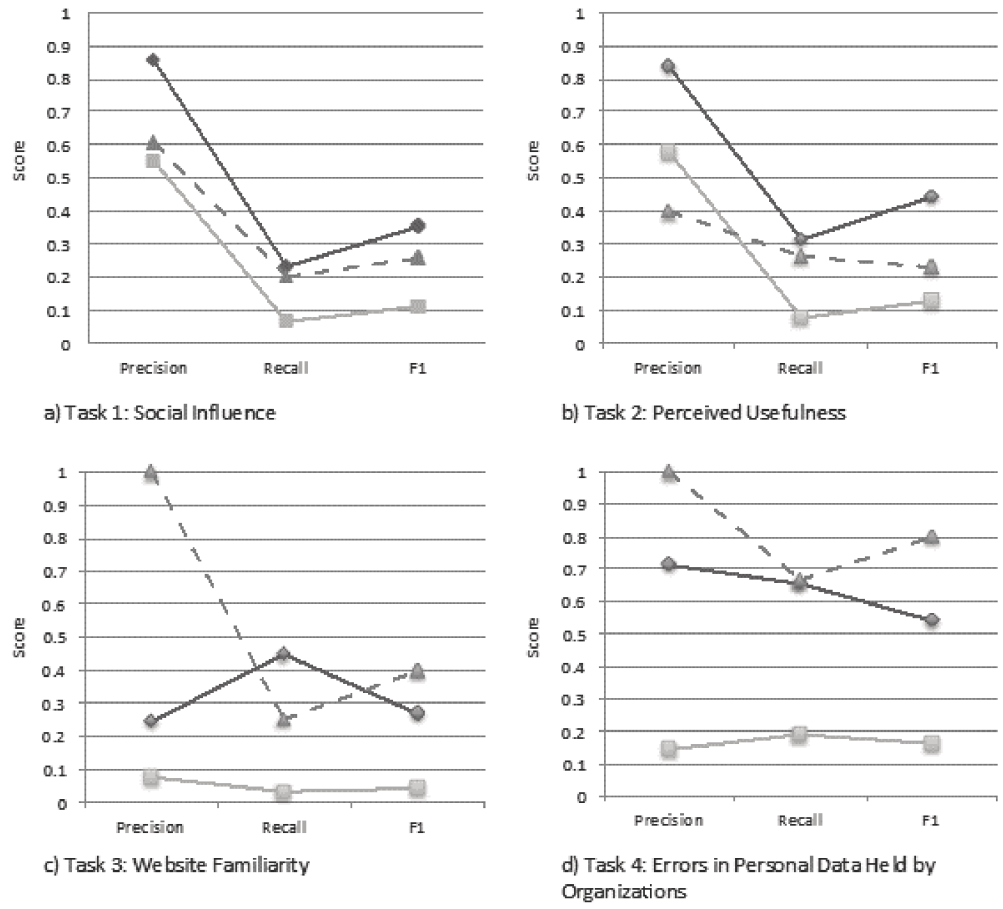


Figure 4. Results (Average Precision, Recall, and F_1-Scores)

responses). For example, if a participant suggested the existence of a *usefulness* construct in an article where past coding had not suggested such a construct, that article was re-examined. There were no disagreements between the two coders for the first half of the sample. As a result, the first author coded the second half of the sample alone. Figure 4 shows results for common and uncommon construct search, and is discussed in the following subsections.

Common Construct Tasks Findings

For the common tasks, the study participants were likely familiar with the literature. Given the attention in IS to (1) the technology acceptance model (TAM; Davis 1989) and the unified theory of acceptance and use of technology (UTAUT; Venkatesh et al. 2003) and (2) *perceived usefulness* and *social influence* in these theories, the participant performance on these tasks suggests a keen understanding of the constituent content of these constructs. When evaluating precision in the *social influence* retrieval task (Figure 4a), participants assigned to the INN search condition correctly recognized the *social influence* construct 86% of the time. While that recognition rate dropped to 55% for users of EBSCO, it should still be considered quite high. The CID_1 algorithm outperformed 10 of the 18 participants in the EBSCO group, with a precision of 60%, but it clearly lagged behind the INN, outperforming only 1 of 18 participants in that group.

As may be seen from Figure 4a, all three approaches—INN keyword (24%), EBSCO (6%), and CID_1 (20%)—performed poorly on the recall evaluation, no approach retrieving even 25% of the known instances of *social influence* in the dataset. Here, CID_1 outperformed 6 of 18 doctoral students using INN and all doctoral students using EBSCO. The F_1-score is shown here in the same order as the independent precision

and recall metrics, with scores of .36, .11, and .26, respectively. For F_1, which represents the overall evaluation for the task, CID_1 outperformed 3 of 18 participants using INN and 17 of 18 participants using EBSCO.

These patterns repeated themselves for the second task, *perceived usefulness,* with the exception that, for precision, the EBSCO experimental group outperformed CID_1, suggesting just how well Ph.D. students know this construct—perhaps the best-known—in the IS discipline. Per Figure 4b, precision for INN was the best of any task at 88%, followed by EBSCO at 59%, and CID_1 at 39%. In this case, CID_1 outperformed no Ph.D. students using INN but 7 of 18 students using EBSCO. The story for recall was quite a bit better for CID_1, and CID_1's performance (26%) was again close to that of the INN group (31%), outperforming 4 of 17 group participants as well as every participant in the EBSCO group (7%) by a wide margin. The surprise here is how hard it is to find all relevant articles containing the desired constructs in a full text search. Finally, the F_1-scores confirm that INN keyword outperformed the other two approaches, .45 versus .13 (EBSCO) and .23 (CID_1). In this case, CID_1 outperformed only one Ph.D. student using INN and 13 of 18 using EBSCO.

Uncommon Construct Tasks Findings

Because of the drive toward novelty, there are many more small, or uncommonly used, construct categories. Most new projects will require the researcher to venture into unknown territory in search of novelty. The third task, *website familiarity* (Figure 4c) provides an example of such territory. For this task, CID_1's precision (100%) outperformed INN (22%) and EBSCO (7%); on recall, CID_1 (24.5%) outperformed EBSCO (3%) but underperformed INN (45%). For this task, CID_1 outperformed all but three INN users and all EBSCO users. For the F_1-score, CID_1 (.40) outperformed both INN (.27) and EBSCO (.04). Here, CID_1 outperformed all 36 human participants.

For the final task, per Figure 4d, for precision, CID_1 (100%) again outperformed both INN (71%) and EBSCO (14%), in the process surpassing 15 of 17 INN users who completed the task and all EBSCO users. For recall, CID_1 (67%) barely surpassed INN (66%) but clearly outperformed EBSCO (19%). For recall, CID_1 was only able to surpass one INN user, but did outperform 14 of 16 EBSCO users who completed the task. For the overall F_1 evaluation, CID_1 (.80) outperformed INN (.54) and EBSCO (.16). CID_1 outperformed 10 of 17 INN users and 15 of 16 EBSCO users. Because design performance will vary based on the construct used as a starting point, Appendix D shares an evaluation of CID_1 with 20 randomly drawn starting points, resulting in further support for CID_1's performance.

We are aware of no other assessments in the literature regarding the human ability to find constructs, and our results indicate poor human capability to locate constructs in full-text. On average, the participants assigned to EBSCO retrieved 9% of the relevant constructs, even in such a small sample of articles. Much has been said about the "reinvention of the wheel" in the world of construct research. This assessment evinces that the problem may be one of human inability to find existing research. We believe the participants in this experiment were quite thorough and motivated to do a good job, as shown by their performance on the large/known tasks, and as such, the experiment only scratches the surface of how incredibly difficult it is to find existing research in an environment plagued by the CIF.

It is encouraging that with this initial algorithm, even experienced doctoral students can be outperformed on some tasks. The CID_1 algorithm performs slightly better than participants using INN keyword search for average F_1 value over the four tasks of .42 versus .40, and outperforms the participants on the unknown tasks. If the goal is high recall, as we believe it should be, there simply are too many constructs now in existence for human beings to be able to play much more of a role than that of creating, training, and evaluating designs. Given that the participants self-reported spending 29.7 minutes per task, the existence of hundreds, if not thousands, more relevant journals would likely degrade human performance appreciably due to eventual satisficing.

Implications of CID_1 for Understanding of Common Constructs: A Meta-Analysis

The preceding results show that CID_1 and INN each perform well in different scenarios, but both appear to have clear benefits over strict reliance by participants searching EBSCO. To understand these results in more detail, we now leave INN aside and examine whether the difference between using CID_1 and EBSCO can actually have a material effect on what researchers can learn by using CID_1 rather than current practices. This section focuses on the case of *common* constructs. We selected the meta-analytic approach (Glass et al. 1981; Stroup et al. 2000) because it enables evaluation the ability of various search processes to produce the best possible results within a corpus. In this case, we conducted the analysis to examine material differences in performance of CID_1 versus full-text search. Based on the gold standard, we found and examined the articles containing both *perceived usefulness*

and *social influence* constructs within the experimental corpus, finding 12 articles that contained at least one pair of these constructs. Upon examination, one article (Karahanna et al. 1999) focused on pre- and post-adoption effects and did not provide relationship statistics for the focal constructs, leaving 11 articles that reported on 29 statistical relationships between the two constructs.

We selected the fixed-effect model (Hedges and Vevea 1998), because we examined all papers available in the experimental time period. We evaluated each participant (as well as CID_1) and his/her average correlation and confidence interval (C.I.) against the population findings, depending on which papers a participant was able to find. The participant was required to discover both constructs in the same paper.

From each paper, we collected the correlation coefficients between the variables of interest—*perceived usefulness* and *social influence*—and sample size. For each correlation coefficient obtained, we calculated the standard error from the sample size as n-3. Using a Fisher's Z-transform, the correlation coefficients were transformed to get r_z, and the fixed correlation coefficient was computed from the transformed value, using a weighted mean average.

We computed the 95% C.I. using the standard error. For each participant, we also (1) calculated the correlation coefficient using just the papers that were found, and then (b) we compared this correlation coefficient to the expected correlation coefficient by noting whether it was within the 95% C.I. obtained for each participant. Table 9 is ordered from best performance to worst performance, and the significance column identifies participants whose answers deviated significantly from the population answer. The first data row in Table 9, entitled *Population*, contains the population calculation using all 29 correlations and their sample sizes, a weighted sample size of 6,843, with an average correlation of 0.303 and 95% C.I. between 0.282 and 0.325.

Per Table 9, only three participants found one or more of the 11 articles (participants 4, 8, and 10), and for one of these, the retrieved articles would have misled him/her into concluding erroneously that the relationship between *perceived usefulness* and *social influence* was significantly different from its population score. By evaluating the z-scores, we see that CID_1 performed between the top two participants with correlation estimates not significantly different from the population and it outperformed the average of the three best participants in spite of having access to only construct items, whereas the participants also had access to construct names, definitions, citations, and the literature reviews inside full-text articles. It should be noted that given the small set of articles found by human participants and algorithm alike, arriving at the right answer in this exercise likely contained elements of luck.

While the precision and recall for this experiment clarified the incumbent need to develop alternatives to full-text search, it was not clear that missing a few papers would make a measurable difference in evaluating relationships between constructs. The meta-analytic approach employed in this article shows that 83% of participants would arrive at the conclusion that the relationship had not been tested at all; of the remaining three participants, only two would arrive at a correct conclusion. If incorrect conclusions and underestimations are possible for well-known construct pairs, such conclusions could be quite common during searches for lesser-known constructs.

Implications of CID for Understanding of Uncommon Constructs: The Error Construct

Common constructs, such as *perceived usefulness* and *social influence*, are crucial to the success of several major theories, including TAM (Davis 1989), UTAUT (Venkatesh et al. 2003), and diffusion of innovations (Rogers 2003), and are, consequently, well known by researchers in the discipline. At times, expert researchers can recall several correspondent construct names. Hence, experts dealing with common constructs sometimes have multiple starting points for their search (for example, a researcher may search for *relative advantage, usefulness,* or *performance expectancy*).

In contrast, researchers searching for uncommon constructs are less likely to know of the existence of past work applying the construct of interest. When they are aware of past work, they are much more likely to proceed from only one starting point. In such cases, one possibility is that the researcher knows that a construct exists, has access to its name, definition, and previously used measurement items, but does not know of related correspondent constructs. Given our findings of low recall for uncommon constructs, what are the behavioral implications to the IS discipline?

Our sample of 1,004 constructs was categorized into 347 distinct construct categories, of which 314 (90.5%) fit our definition of uncommon—defined here as occurring five or fewer times in our dataset. The large proportion of uncommon constructs could be seen as indicative of the IS field's focus on theoretical extensions of existing theories. Uncommon constructs serve a crucial function in validating common constructs (Cronbach and Meehl 1955) and to better understanding what drives behavior. Common constructs are sometimes extended in thousands of articles (Larsen et al. 2014), yielding many new constructs, the overwhelming majority of which remain uncommon.

Table 9. Meta-Analytic Evaluation of Participant and Algorithm Performance

	Each Individual Meta-Analysis				Comparison Versus Population		
Name	Articles Found	~Sample Size	Correlation	Low [95% C.I.]	High [95% C.I.]	z-score	Significant Difference
Population	*11*	*6,843*	*0.303*	*0.282*	*0.325*	n/a	n/a
Best participant	2	957	0.306	0.248	0.363	0.095	
CID	**1**	**1,335**	**0.268**	**0.217**	**0.317**	**1.292**	
Second best participant	1	339	0.370	0.275	0.458	1.352	
Third best participant	3	1,743	0.379	0.339	0.419	3.216	***
Participants 4 - 18	0	—					

Results significantly different from population answer. *p < .05, **p < .01, ***p < .001.

The small corpus used in our experiment should have compensated for the lower motivation likely in an experimental setting. Therefore, our experimental findings on low expert ability to find constructs in full-text articles were surprising. If indeed researchers are able to find only a small proportion of the relevant construct articles, what are the implications for research practice? *Ceteris paribus*, missed constructs will disproportionately impede domain sampling and content validity (Churchill 1979) for uncommon constructs relative to common constructs.

We focus here on the specific case wherein recall under the full-text search condition (EBSCO) was at its peak, 19% (relative to 3%, 6%, and 7%). This experimental task, to find the construct named *errors [in personal data held by organizations]*—is a privacy-focused construct associated with items such as "Companies should take more steps to make sure that the personal information in their files is accurate." This was where both CID_1 and all the participants (regardless of system use) performed the best for recall. When evaluating the reason for this performance, we found this task, relative to all others, to be easy. Specifically, the experimental task was to find three identically named constructs with almost identical measurement items on the topic of privacy, a context that was quite unique within the sample of articles. This was a case where experimental participants had a great advantage over CID_1, in that the three articles containing this construct were heavily connected through citations, and no knowledge of correspondent construct names was needed.

The *errors* construct was first proposed by Smith et al. (1996) in an article cited no less than 20 times in Stewart and Segars (2002) and 21 times in Malhotra et al. (2004). Malhotra et al. (2004) also cited Stewart and Segars (2002) 11 times. While it is hard to imagine any researcher missing these citations when sitting down to read the found articles, the 19% recall should be taken as an indication of how many articles remain unfound in this initial step of the review process. Such an implication must also take into account that, in a setting of hundreds of journals and as many relevant conferences, examination of citation links in the articles found is only possible to the extent that the number of retrieved articles is below the threshold set by a researcher's available time. Even for uncommon constructs such as *errors*, and even if the precision could be perfect, the number of papers found in a search may consist of dozens or even hundreds of articles, each with dozens of citations to explore.

Smith et al. proposed an information privacy instrument measuring the "primary dimensions of individuals' concern about organizational information privacy practices." The resulting instrument contained a first-order model containing four constructs; *collection, errors, unauthorized secondary use,* and *improper access*. These constructs focus on attitudes toward company collection of private data, the existence of errors in the resulting company database, company use of data for unauthorized purposes, and access to the data by unauthorized individuals, respectively. Accessing this paper provides the researcher with the knowledge that the *error* construct exists and also provides understanding of constructs closely related to *error*. However, without finding Stewart and Segars, the researcher will not know the four constructs may be presented as a second-order model, which thereby provides stronger evidence of the relationship between the second-order model, *computer anxiety,* and *behavioral intention*. Malhotra et al. (2004) reimagined *privacy concern* after examining the *error* construct along with other constructs. They suggested that the *error* construct was no longer necessary when privacy is recontextualized within social contract theory (Donaldson and Dunfee 1994). Malhotra et al.'s path values suggested that further examination was necessary, but they introduced *trusting beliefs, risk beliefs, information sensitivity,* and additional evidence for relationships to *behavioral intention*, all relationships that would be missed by the vast majority of the EBSCO participants.

Swanson (1986) convincingly showed that academia's divide-and-conquer strategy (into academic fields), necessitated by the increasing complexity and sheer mass of research, at times left even seemingly obvious or "public" knowledge undiscovered. He suggested that most units into which we split academia are undoubtedly connected to other units, that there are far more potential relationships between units than there are units themselves, and that the system of organization is not organized to cope with combinations. Our experiment shows that even within the narrowest definitions of such units (two highly selective journals in one relatively narrow field), existing relationships are likely lost during literature reviews due to the complexities of full-text search. In essence, the experimental participants would have missed most of the nomological network tied to the *error* construct. We provide evidence that the search for knowledge is more complex than suggested by Swanson. While Swanson addressed undiscovered knowledge, we found the vast majority of *discovered* knowledge is hidden from the individual researcher by the very nature of the search process itself.

Discussion

Researchers are not confined to gathering empirical evidence in order to support hypotheses but can also substantiate hypotheses by fitting them to existing theories (Kaplan 1964). When hypothesizing a relationship between constructs A and B, detecting constructs A' and B' in another theory arguably provides as much evidence for the relationship as a new test of the relationship (Bacharach 1989). Similarly, in meta-analysis, finding a published relationship between X' and dependent variable Y can be considered a conceptual replication of the relationship between X and Y (Hunter and Schmidt 2004, p. 436). As long as the constructs reflect the same phenomenon, such analysis arguably improves construct validity (Hunter and Schmidt 2004, p. 448). Given the enormous corpora of published research, attention should begin to shift from generation of new research to integration of past research and the tools necessary to detect correspondent relationships.

We confirm that NLP-based designs can indeed address the construct identity fallacy, including its jingle and jangle components. Such designs can support meta-analysis, reviews, and construct validity evaluations. In addition to this major implication, the work has several secondary findings. First, we establish that construct renaming reduces construct recall, especially during full-text search. Second, we show the relative frequencies of jingle and jangle fallacies in a sample of IS journals. Third, extending Swanson's (1986) conclusions, we demonstrate that relationships need not be hidden in order to be inaccessible to researchers. Fourth, while use of INN in many cases outperformed CID_1, INN still requires the active involvement of human experts, whereas CID tools can examine large sets of constructs automatically. Further, CID should be relatively invariant in terms of reflective and formative constructs (see the second section of Appendix C).

The construct identity fallacy represents an enormous challenge for researchers laboring to connect their research to the past and incrementally build a future for their discipline. During evaluation of designs, CID_1 was found to be most efficacious for addressing overall CIF and the jangle fallacy, and second best for the jingle fallacy, after LSA. Human participants using INN during the common construct tasks outperformed CID_1, but CID_1 outperformed participants using INN and those using EBSCO full-text search during uncommon construct tasks. When applying a meta-analytic evaluation of the results for the common construct tasks in which participants outperformed CID_1 when evaluated against the gold standard, we found that CID_1 outperformed the vast majority of participants using full-text search and that most participants would erroneously conclude that no past research had addressed the relationship between the two constructs.

By selecting only constructs where three or more items were available, one-item and two-item constructs were excluded, leaving an open question about how well the tool would work for such constructs. While one- and two-item constructs may not be as common as three- or more-item constructs, the issue may be pervasive because authors often include only one or two sample items for each construct in the published paper. This case, however, is no different from that faced by researchers doing factor-analytic research, as items must be procured before analysis is possible.

Our experiment and meta-analytic evaluation show not only the importance of such research due to the herculean task of finding constructs using full-text search, but also the usefulness of NLP-based designs in detecting and integrating constructs based on linguistic features. In fact, the experimental meta-analysis showed that the *majority* of Ph.D. student participants retrieved results that would have resulted in incorrect conclusions about the relationships between constructs. This concealment of past research discloses a worrisome question: What is the value of research if only a fraction of past research is accessible to any individual researcher when needed? We suggest here that because such knowledge is available in the plain text of papers, ontology learning approaches are necessary in order for researchers to access a larger proportion of our shared knowledge heritage.

Conclusion

The construct identity fallacy has existed for over 100 years across a number of behavioral and social disciplines with no viable large-scale solution hitherto proposed. CIF has major implications for the retrieval of relevant research. In evaluations of participants' recall performance in full-text search, we found that participants could detect relevant articles containing a specific construct on average only 9% of the time, and relevant articles containing a pair of common constructs on average only 3% of the time. The experiment also found that while Ph.D. student participants were able to recognize familiar constructs when these constructs were already extracted from the full-text, this ability decreased markedly when participants were asked to detect and delineate those constructs from a corpus of full-text papers. The construct identity detector tool presents a technical solution to CIF that diverges from existing approaches because it does not depend on survey respondents, and therefore allows large-scale investigation of constructs. CID could enable science to function better by encouraging creative research unrestrained from the need for naming conventions, while simultaneously identifying construct identity fallacies, understanding their effects, and allowing improved research integration.

This work is part of a larger Design Science research program designed to address construct identity fallacies, and it takes a step toward providing the construct identity detection capabilities necessary to make ontology learning possible and useful through automatic creation of construct knowledge bases, as proposed by Li and Larsen (2011, 2013). Our first attempt to evaluate the various designs provides a direction for further exploration and refinement of a design that can help identify correspondent constructs. While creation of construct knowledge bases will require work, the tool outlined in this article is notable because items from thousands of constructs may be evaluated, and construct pairs with high similarity scores are more likely to be correspondent. Finding additional relationships between constructs are shown to have major behavioral implications for our understanding of constructs and real-world phenomena.

Finding a shared construct in two theories may enable the connection of those two theories. While the concept of fitting hypotheses to existing theories is deceptively simple, the successful implementation of this concept has the potential to change the conduct of research in the behavioral sciences. Success in creating technologies able to address the construct identity puzzle for all the behavioral disciplines—as this article found possible for the IS discipline—has the potential for discipline-wide transformation across the social and behavioral sciences.

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MIS Quarterly

METHODS ARTICLE

A TOOL FOR ADDRESSING CONSTRUCT IDENTITY IN LITERATURE REVIEWS AND META-ANALYSES

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Appendix A

Natural Language Processing Designs

This appendix provides background details on the NLP designs used in our paper. For each, the steps necessary for implementation are outlined.

Latent Semantic Analysis

The underlying idea of latent semantic analysis (LSA) (Deerwester et al. 1990) is that the aggregate of all of the contexts in which a given word does and does not appear provides a set of constraints that determines the similarity of meanings for words, and sets of words, to each other (Landauer et al. 1998). Thus, when two terms occur in contexts of similar meaning, even if they never occur in the same passage, LSA represents them as having similar meanings.

LSA Steps

Step 1. Preparing term-document matrix. For the LSA process, we employed minimal pre-processing. Starting with the paragraphs from the *MIS Quarterly* and *Information Systems Research* articles that contained the sample of constructs, we removed non-alphanumeric characters and converted all words to lower case; we then stemmed the words using the Porter (1980) algorithm and weighted the term-document matrix using log-entropy and normalization.

Step 2. Creating semantic space. After weighting and normalization, the matrix A is decomposed using singular value decomposition (SVD), a mathematical algorithm similar to a factor analysis, with the result being a *semantic space* containing term vectors, each infused with an "understanding" of the term.

Step 3. Projecting items into the semantic space. Given the query q, which is a construct item, a query vector $\vec{q}$ is obtained through an aggregation of term vectors relevant to the item. In our research, we project every item into the semantic space as a query vector $\vec{q}$, and that vector is saved as $\vec{q}_n$ for future item-item analysis, where n is the total number of items. This improves speed of the analytics and enables separately stored solutions for search-engine purposes.

Step 4. Finding similar items. To find similar items to $\vec{q}$, the query vector is compared against all the items stored inside the semantic space, $\vec{q}_n$, using the cosine similarity measurement. We deem vectors in $\vec{q}_n$ that yield the highest cosine scores corresponding to $\vec{q}$ to be more semantically relevant; we deem those with low cosine scores to be relatively irrelevant.

Semantic Space Selection

"The selection of dimensionality in LSA is fraught with problems" (Kakkonen et al. 2008, p. 280), and in fact, researchers have yet to find a generalizable solution to dimensionality selection (see, e.g., Bingham and Mannila 2001; Globerson and Tishby 2003; Landauer et al. 1998). In IS, this issue was prominently featured in a very thorough LSA exposition by Siderova et al. (2008), where domain-specific 5-, 13-, and 100-dimensional solutions were proposed and analyzed. While research has suggested that 300-dimensional solutions work well (Landauer and Dumais 1997), that result was not based in theory and is not likely to generalize perfectly into the psychometric domain. The purpose of our article is to show that natural language processing works to address the construct correspondence problem rather than to fully optimize such designs. This work was simply conducted using a 300-dimensional solution. Our own examinations have shown that when used for construct items, LSA semantic spaces between 200 and 500 are roughly equivalent, but higher dimensions do marginally better (although with associated increases in processing time).

Latent Dirichlet Allocation

We translated all words in each item into k topical probabilities according to the posterior distribution. Hence, each item was depicted as a k-dimensional vector representing its topic distribution. To compute the similarity of the items, we computed the cosine of the item vector. The following list of steps illustrates the process.

Step 1. Building the LDA model. We used the same paragraphs used for LSA to build the LDA model.

Step 2. Preprocessing. We preprocessed paragraphs by removing stop words,[1] punctuation, and non-alphabet characters.

Step 3. Computing topic distribution. An advantage of online LDA is its ability (1) to process the whole document collection in a single, full pass; (2) to converge rapidly; and (3) to yield an accurate topic estimation. In this step, we defined k to present the number of topics. We set k to 300 after examination of values from 50 to 500 in increments of 50. That examination follows in the next section.

Step 4. Building an item subspace. Once the LDA model was ready, we projected each item into the model and transformed it into the k-topic distribution format. Similar to LSA, this represented the model subspace for all of the items.

Step 5. Finding similar items. It is our assumption that two items are semantically similar if they share identical topic distribution. Thus, we defined the similarity as the angle cosine of the two item vectors in the subspace.

Topical Probabilities

We evaluated 10 LDA designs using different numbers of topical probabilities, where k = (200, 250, 300, 350, 400, 450, 500). Improvements are apparent when increasing k from 50 to 300, with the greatest improvements seen in the change from 150 to 200. However, increasing k beyond 300 yielded only insignificant improvements along with longer convergence time. A close examination revealed that the k values around 300 were able to transform marginal false negatives to true positives and eliminate a high number of false positives. LDA should work well theoretically for the problem at hand, but the nature of the texts used to measure a construct (the construct measurement items) is quite different from the majority of paragraphs in an academic paper. In addition, we are aware of no other research that uses LDA in quite this way. Unlike LSA, the representation of items in LDA is not able to take into account synonymy effectively. While LDA may be better fit for detecting latent topics, our experiments show that LDA did not perform well for establishing similarity. One of the reasons for this discrepancy is the representation of topic distribution, as the short vectors may diminish its performance during calculation of similarity. For example, LDA similarity for "task" and "job" yielded 0.00, whereas in LSA, it yielded over 0.9.

[1]We used the University of Glasgow stop word list (http://ir.dcs.gla.ac.uk/resources/linguistic_utils/stop_words). A pdf copy of this list is available from the authors for archival purposes.

Knowledge-Based Approaches: LI

The semantic similarity of item pairs is computed by taking the path length and the depth of two words in a hierarchical semantic knowledge base, i.e., WordNet (Miller 1995; Poli et al. 2010). WordNet is a hierarchical lexical reference system consisting of "English nouns, verbs, and adjectives…organized into synonym sets, each representing one underlying lexical concept" (Miller et al. 1993, p. 235). Later work added adjectives to WordNet. LI relies on WordNet to measure word similarity because it is one of the richest and most accurate lexical dictionaries ever crafted, correlates well with human judges (Miller 1995), is readily available, and does not adhere to a specific domain.

Word-order similarity is factored in, turning each sentence into a vector by ordering the words as they appear and computing the difference of the word orders. Finally, in order to separate the informative words from those that are not, information content of each word is normalized onto each similarity score from the Brown University Standard Corpus of Present-Day American English (Marcus et al. 1993).

Knowledge-Based Approaches: MI

Building on WordNet, MI's (Mihalcea et al. 2006) word specificity refers to the specific meaning of words (e.g., collie and sheepdog) versus generic concept words (e.g., animal and mammal). The similarity measure gives higher weight to precise, specific-meaning words than to abstract, generic-concept words. Word specificity is computed using the inverse document frequency (IDF) algorithm (Jones 1986), which was applied to the British National Corpus. IDF assumes that rare words have greater discriminatory weight than common words. Additionally, IDF similarity scores are normalized to reflect word specificity in the sentence. MI averages word-similarity metrics from six different sources: Jiang and Conrath (1997), Leacock and Chodorow (1998), Lesk (1986), Lin (1998), Resnik (1995), and Wu and Palmer (1994). These metrics were created to measure word relatedness and similarity by calculating the shortest distance between given words' synsets (sets of synonymous words) in the WordNet hierarchy; the shorter the distance between words, the higher the similarity score.

MI was designed to measure short text similarity and has been used at the sentence level by Mihalcea et al. (2006). Like LI, the MI similarity score is a number between 0 and 1, where 0 indicates no semantic overlap and 1 indicates exact match. The MI sentence similarity measure is computed for two candidate sentences, S_1 and S_2 as follows:

Step 1. Identifying part-of-speech (POS). The process begins with tokenization and part-of-speech tagging of all the words in the sentence into their respective word classes (noun, verb, adverb, and adjective, as well as cardinal numbers).

Step 2. Calculating word similarity. Each word in the sentence is evaluated against all the words from the other sentence to find the highest semantic similarity (*maxSim*) from the six word similarity metrics: Jiang and Conrath (1997), Leacock and Chodorow (1998), Lesk (1986), Lin (1998), Resnik (1995), and Wu and Palmer (1994). These metrics were originally created to measure concept likeness rather than word likeness, but they are adapted in our study to compute word similarity, by computing the shortest distance between given words' synsets in the WordNet hierarchy. The word–word similarity is computed only on the words from the same word class, which are either from noun or verb word classes, because WordNet contains separate semantic trees for nouns and verbs; it is thus not possible to obtain similarity between nouns and verbs using WordNet distance. For other word classes such as adverb, adjective, cardinal, and unknown words, whole word matching is used instead. The word-word similarity measure is directional. It begins with each word in S_1 being computed against each word in S_2 and then vice versa.

Step 3. Calculating sentence similarity. Once the highest semantic similarity (*maxSim*) for each word in the sentences is computed, it is normalized with inverse document frequency (IDF) to weight rare and common terms. The normalized scores are then totaled for a sentence similarity score, Sim_{MI}, as follows:

$$Sim_{MI}(S_1, S_2) = \frac{1}{2} \times \left(\frac{\sum_{(w \in S_1)} maxSim(w, S_2) \times IDF(w)}{\sum_{(w \in S_1)} IDF(w)} + \frac{\sum_{(w \in S_2)} maxSim(w, S_1) \times IDF(w)}{\sum_{(w \in S_2)} IDF(w)} \right) \quad (1)$$

where $maxSim(w, S_2)$ is the score of the most similar word in S_2 to w and IDF (w) is the inverse document frequency of word w.

The CID Hybrid Approaches: The CID_1 Model

The process steps for calculating CID_1 similarities of two given items are as follows:

Step 1. Forming both items into a joint text. The comparing item and the joint item are formed into a matrix to allow each word from different items to be compared. Table A1 shows that process for the items, "RAM keeps things" and "the CPU uses RAM." The word-word similarity here is computed using LSA cosine similarity scores.

Step 2. Identifying the maximum similarity score. For each column (the columns represent the element of the lexical word vector), the maximum similarity score is identified, which, in turn, is normalized using information content from the Brown Corpus.

Step 3. Forming a lexical word vector for the first item. Only the similarity scores above the preset threshold of 0.2 (developed for LI) are selected to form a lexical word vector. Scores below the threshold are set to zero, based on empirical findings that such words are too dissimilar to score (Li et al. 2006). In this study, we did not test whether a different threshold from LI's 0.2 would work better for CID_1. Extensive differences between the LI and LSA similarity measures suggest that testing alternative thresholds might improve the CID_1 algorithm, but we did not expect it to change the outcome of the evaluations. Thus the lexical word vector for the first item is $\vec{S}_1$ = [0.39, 0.33, 0.179, 0.00, 0.074, 0.23].

Step 4. Forming a lexical word vector for the second item. The step is identical to step 3, but using Table A2. This yields $\vec{S}_2$ = [0.19, 0.00, 0.16, 0.03, 0.389, 0.04].

Step 5: Deriving the word-order vectors using joint word. The word-order vectors are also derived using joint word, such that words in the vector are assigned with unique indices according to the order in which they appear in candidate items. For example, the word-order vector for the joint words "*RAM keeps things The CPU uses* " is [1,2,3,4,5,6].

To derive the word-order vector for "RAM keeps things," we assign the word-order index to the corresponding words in the candidate item. The first word "RAM" has an index of 1, and "keeps" has 2, etc. However, since the item does not have the words "The," "CPU," and "uses," the index will be the most similar word from the index that is computed in Step 2. For example, according to Table D1, "The" is similar to no other word and thus it has an index value of 0, whereas "CPU" and "uses" are most similar to "things," which has an index of 3. Thus the word-order vector for the first item is $\vec{O}_1 = [1, 2, 3, 0, 3, 3]$. The same process applied to the second item yields, $\vec{O}_2 = [1, 0, 3, 4, 5, 6]$. The minimum similarity threshold also applies here. For any word that has similarity less than the preset threshold, the value is zero. Likewise, the other comparing item is also derived using the same approach.

Step 6. Computing the similarity of word-order information to distinguish the meaning of two items attributed to syntactic differences. The word-order similarity score is then computed as follows:

$$Sim_{order}(S_1, S_2) = 1 - \frac{\|\vec{O}_1 - \vec{O}_2\|}{|\vec{O}_1 + \vec{O}_2\|} \tag{2}$$

Step 7. Obtaining items' similarity score. Similarity scores are obtained by calculating the cosine angle of the lexical word vectors,

$$Sim_{semantic}(S_1, S_2) = \frac{\vec{S}_1, \vec{S}_2}{\|\vec{S}_1\|, \|\vec{S}_2\|} \tag{3}$$

Step 8. Calculating overall sentence similarity score. Finally, the overall item similarity is defined as the combination of lexical semantic similarity and word-order similarity, which are operationalized in in the following:

$$Sim_{Li}(S_1, S_2) = \gamma\, Sim_{semantic}(S_1, S_2) + (1 - \gamma) Sim_{order} \tag{4}$$

where γ is a relative contribution of semantic similarity and word-order similarity to the overall similarity. According to Li et al, γ is a value greater than 0.5.

Table A1. Deriving Semantic Vector for Item 1 (S_1)						
S_1	RAM	keeps	things	the	CPU	uses
RAM	1	0.05	0	0	0	0
keeps	0.15	1	0	0	0	0
things	0.33	0.39	1	0	0.34	0.83
Sim	1	1	1	0	0.34	0.83
Weight	I(RAM) I(RAM)	I(keeps) I(keeps)	I(things) I(things)		I(CPU) I(things)	I(uses) I(things)
$\vec{S}_1$	0.390	0.330	0.179	0	0.074	0.23

Table A2. Deriving Semantic Vector for Item 2 (S_2)						
S_1	RAM	keeps	things	the	CPU	uses
The	0	0	0	1	0	0
CPU	0	0	0	0	1	0
uses	0	0	0.25	0	0	1
RAM	1	0	0	0	0.33	0
Sim	1	0	0.25	1	1	1
Weight	I(RAM) I(RAM)		I(things) I(uses)	I(The) I(The)	I(CPU) I(RAM)	I(uses) I(uses)
$\vec{S}_2$	0.190	0	0.16	0.03	0.389	0.04

The CID Hybrid Approaches: The CID_2 Model

While based on the MI sentence-similarity algorithm, CID_2 does not perform part-of-speech (POS) tagging, and word similarity is directly calculated using LSA. Similarity scores are then normalized with inverse document frequency (IDF) to reflect rare and common terms (Mihalcea et al. 2006). The following steps illustrate the sentence-similarity score process developed for CID_2.

Step 1. Computing word similarity using LSA. The process begins directly with word-word similarity. Given two words, the cosine similarity is calculated based on the LSA semantic space.

Step 2. Computing sentence similarity. Once the word-similarity scores are obtained for each sentence, they are normalized with correspondent IDF, and the sentence-similarity scores are computed by replacing the original equation with the LSA similarity measure, which yields the following equation for CID_2:

$$CID2(S_1, S_2) = \frac{1}{2} \times \frac{\sum_{(w \in S_1)} Sim_LSA(w, S_2) \times IDF(w)}{\sum_{(w \in S_1)} IDF(w)} + \frac{\sum_{(w \in S_2)} Sim_LSA(w, S_1) \times IDF(w)}{\sum_{(w \in S_2)} IDF(w)} \quad (5)$$

While we had expected CID_2 to outperform LSA, careful consideration of the evidence suggests a good reason for CID_2's performance. We examined a number of variables using two separate linear regressions on the relative ranking of item pairs for CID_2 versus MI, split on whether the item pairs were correctly or incorrectly classified. In both analyses, the max(IDF) for each item pair emerged as the most important negative influence on CID_2 design performance. In a separate experiment, we removed the IDF weighting from CID_2 and compared its performance against all designs on a sample set of item pairs, which resulted in CID_2 moving up from the bottom to the third best performer (just behind CID_1 and LSA, respectively). Detailed examination also indicated that the weighting of IDF in CID_2 did not scale well with LSA (between common and informative words) because the geometrical scaling of word occurrences is less meaningful in LSA; the virtue of its similarity scores is maneuvered through algebraic operations in the semantic space. CID_2 was not included in the final CID tool in lieu of LI, because LI serves the function of providing a WordNet knowledge-based design for the final tool, which is important for future contexts of use.

Appendix B

Overview of Gold Standard

Data collection was performed by a team assembled for a larger project funded by the National Science Foundation. The team consisted of advanced undergraduate and master's level graduate students at a large, research university in the western United States. Students were selected for interviews based on previous research experience, GPA, academic awards, and interest in research. Selected applicants received a two-hour training session. This training was followed by an hour-long construct-extraction task consisting of the extraction of all variables, definitions, and items from an academic article. Student performance on the task was then graded against a gold standard of previously extracted variables, definitions, and items, and validated by our Chief Senior Research Assistant. The top scorers were hired as research assistants (RAs), resulting in the top 3 percent of applicants accepting positions. Hired RAs received an additional 10 hours of training along with careful monitoring of their first 100 hours of extraction. After 200 to 300 hours of extraction experience, the best 20 to 30 percent of RAs were promoted to senior RAs. All articles were first extracted by a junior or senior RA and then audited by a senior RA.

Categorization Process

Categorization was carried out by a research team, which consisted of one experienced faculty researcher, three doctoral students, and four experienced research assistants (RAs with at least 500 hours of experience in construct extraction).

Overall, the categorization task was divided into two stages: rough categorization and refined categorization. Rough categorization involved grouping constructs with highly similar properties into the same cluster to make the task cognitively tractable. The refined categorization process resulted in hierarchies, sub-hierarchies, and categories of the groups generated by rough categorization. The refined categorization process began by labeling the hierarchical clusters. The taxonomy in this study was restricted to two hierarchical levels. For example, the main category, "Trust," had the subcategories, "Trust in Benevolence," "Credibility," and "General Trust." The hierarchical structure was used solely to reduce the cognitive load on the participants. Ultimately, the only important consideration was whether a pair of constructs should be placed in the same category (correspondence) or different categories (independence).

Once each construct was assigned to one of the categories, annotator teams took turns examining all categories to ensure that constructs had similar contexts and fit the definition of correspondence. Some clusters—usually those with a high number of constructs—required several examinations by different teams. Each category was examined by at least two teams, with the team containing the faculty member always doing the final examination. If any team did not fully agree with the previous team, that team would reorder and move constructs, and a third team would reexamine the category. Constructs were assigned only to one category. Both the rough and refined categorization tasks were carried out in stages across six days, and they consumed about 200 person-hours.

Evaluation

Finally, in recognition of the complexity of constructing a taxonomy for such an intricate dataset, two researchers, having a combined 15+ years of research experience and no exposure to the taxonomy process, were chosen to evaluate the dataset. The experts were given a semi-randomized set of 300 construct pairs, where each pair had a 0.5 probability of being either correspondent or independent. The inter-rater agreements of correspondence versus independence between the two experts and the taxonomy were at 85 percent and 90 percent. The resulting Cohen's (1960) kappas were 0.68 and 0.79, indicating agreement levels that are considered "substantial" and close to "almost perfect" by Landis and Koch (1977).

The Taxonomy

We share this taxonomy with the caveat that it represents only a partial view of constructs in the two journals *MIS Quarterly* and *Information Systems Research*. It consists of constructs reported with a minimum of three items during the period from 1983 to 2009. The taxonomy is irrelevant to the construct-identity-detection designs in this article, as the designs operated at the lowest level of the taxonomy and did not consider the hierarchical structure of the taxonomy. However, the hierarchical structure was invaluable in reducing the cognitive strain of the categorization exercise reported in the evaluation section.

In the case of the present taxonomy, a bottom-up process was employed, which tends to guide the taxonomy toward the larger topics. In other words, initial groups such as *adoption* and *development* will tend to grow faster and attract constructs earlier, before other established groups have a chance to spawn on their own. These originally smaller groups will become even smaller hierarchies as their remaining constructs are organized. For example, constructs related to top management involvement, which were categorized into the *IS development* hierarchy, may have ended up in the *leadership* hierarchy in other categorization processes where such seemingly related constructs as leader/subordinate relationships exists. Further, the lean hierarchy named *academic* could have attracted more constructs under other circumstances as well as a larger hierarchical structure if a classification exercise had been undertaken with a preexisting top-level structure. Nevertheless, any topic of reasonable size is likely to be represented as a hierarchy, and the hierarchies and their construct categories represent a potentially useful step toward the development of an ontology in which such construct categories would need to be recreated with links to related construct categories. With these caveats, we describe here the taxonomy of IS constructs created for this project.

By size, the hierarchies of the taxonomy are *IT adoption* (412 constructs), *IS development* (102), *trust* (63), *information/data* (48), *task/job* (45), *interorganizational* (43), *IS function* (43), *communications* (40), *organizational* (39), *learning* (23), *purchase* (22), *group* (19), *knowledge* (17), *judgment and decision making* (13), *leadership* (10), *strategy* (10), *general psychology* (7), *ethics/morals* (6), and *academics* (4).

That the IS field is focused on *IS adoption* (41%)—perhaps unsurprisingly to anyone involved in IS research—is among the most immediate findings. Within the adoption hierarchy, the largest sub-hierarchy by far is the one we termed *technology factors*, constructs that focus on user perceptions about technology features such as *ease of use* and *usefulness*. The second largest sub-hierarchy, which could have fit in another taxonomy under *technology factors,* was named *affective factors,* which focuses on user affect and attitude, including immersion in a technology. The third largest, *use factors*, focuses on constructs related to amount and extent of use, intended or self-reported. The last sub-hierarchies, *social/external factors* and *efficacy factors*, round out the hierarchy with a focus on facilitating conditions and technology self-efficacy.

The second largest hierarchy, *IS development* (10.2%), focuses on the process of developing information systems, primarily *participation/support*, which is about (1) the perceived level of user participation during the process and (2) (top-)management involvement in the same. *Process methodology* includes methodology, along with risk factors such as requirement-focused constructs. Less obvious is the focus on *trust* (6.3%), perhaps suggesting a level of uncertainty or even insecurity about the interface between technology and users, as well as between technology users and other individuals/organizations mediated through technology. The *information/data* (4.7%) hierarchy speaks to a focus on the data and their transformation process, with quality and understanding representing key foci. Other hierarchies of interest included *communication*, *privacy*, *learning,* and *purchase*. Smaller hierarchies are not discussed in this analysis.

What stands out in this gold standard is the great variety in level of analysis within the construct set (Burton-Jones and Gallivan 2007). The majority of research is plainly at the individual level (*IS adoption* being a typical example), but we found research directed at the *task/job* (4.5%), *group* (1.9%), *IS function* (4.3%), *organization* (3.9%), and *interorganizational (4.3%)* levels of analysis. In Table B1, we share the whole set of hierarchies. Readers are warned against the spurious belief that construct items and definitions of constructs are predictable based on construct, category, sub-hierarchy, and hierarchy names alone.

Table B1. The Gold Standard Taxonomy

Hierarchy	Sub-hierarchy	Category	# Constructs
Academic		Masculinity/femininity values	1
		Revising activities	1
		Writing importance for tenure/promotion	2
Communications	Amount of communications	Individual Information quality	2
	Channel characteristics	Communication channel characteristics	4
		Virtual co-presence	1
	Communication knowledge	Knowledge sharing to/with groups outside of the organization	6
		Knowledge sharing obligation	1
		Loss of knowledge power	1
	Communications psychology	Conflict	4
		Self-worth from knowledge sharing	1
	Communications relationship	Anticipated reciprocal relationships through knowledge sharing	1
		Expectation of reciprocity in knowledge sharing	1
		Shared understanding of role within organization	2
		Strength of ties	1
	Communication satisfaction/rewards	Economic incentive	1
	Communications quality	Communication quality	7
		Interaction quality	1
		Quality of communication interface	1
	Etc.	Communication ease	1
		Communication seeking and reception	1
		Experimental similarity	1
		Governmental contention	1
		Vertical coordination	1
Ethics/morals		Ethical behavior	1
		Internet ecology	1
		Moral intention	3
		Predisposition to justice	1
General psychology	Negative	Negative affectivity	2
		Trait anxiety	1
	Neutral	Effort to change mental model	1
		Long-term orientation	1
	Positive	Need for cognition	1
		Self-monitoring	1
Group		Control culture	1
		Dynamic capabilities	1
		Employee versus job orientation culture	2
		Group cohesion	7
		Inclusivity culture	1
		Innovative culture	2
		Normative versus pragmatic work norms	1
		Perceived outcome quality	1
		Relational capital	1
		Satisfaction with group	1
		Team member accountability	1

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
Information/Data		Accessibility	1
		Argument quality persuasiveness	2
		Compatibility	4
		Completeness of information and breadth of information exchange	4
		Currency and availability of information	2
		Data collection	1
		Information quality	9
		Information quality satisfactory	3
		Information quality specific yet broad satisfaction	3
		Information quality importance	2
		Information reliability	4
		Information reliability importance	1
		Information that meets needs	2
		Information understanding	5
		Information usefulness	1
		Locatability	2
		Understandability and reliability	2
Interorganizational	Extent/strength of collaboration/relationship	Collective sanctions	1
		Cooperative norms expectations to work together	1
		Degree of IS outsourcing	2
		Interorganizational process integration	6
		Mutual adoption	1
		Supplier responsiveness to customer needs	2
	Market factors	Asset specificity	3
		Contracting flexibility	1
		Coordination costs between firm & vendor	2
		Cost/benefit of external service provider	1
		Exchange safeguards	1
		Interorganizational relationship standards	1
		Interorganizational relationship efficiency	1
		Network effectiveness	1
		Supplier choice	1
		Transaction cost	1
	Psychological factors	Motivation to comply in general	1
		Persistent expectation of subordinance due to previous contract	1
		Relationship satisfaction	1
	Success	Effectiveness of product development work unit	1
		Relationship performance benefits from interaction	2
	Task/process factors	Interorganizational process modularity	1
		Trading partner readiness	1
	Etc.	Customer obligation	1
		Interorganizational pressures to adopt technology	2
		Obligation for accurate project scoping	1
		Obligation for building effective team	1
		Obligation for clear authority structure	1
		Obligation for taking charge	1
		Relationship management skills	1
		Restricted network access for strategic purposes	1

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
IS development	Participation/support	Perceived user participation	10
		Breadth of involvement in planning	1
		Client formal control over vendor	1
		Desire to participate in process/group	1
		Functional capability of the application service provider	1
		Management involvement in planning	7
		Obligation to ensure top management support	1
		Organizational support	1
		Perceived provider performance	1
		Planning cooperation	1
		Technical service guarantees	1
		Top management involvement - quality policy and goals	1
		Top management technology attitudes	3
	Process methodology	Amount of SDM use	2
		Barriers to implementing the methodology	1
		Frequency of use of coordination technology	1
		Methodology provides an effective planning process SDM	5
		Methodology provides a high-quality end product SDM	4
		Process standardization	3
		Quality of interactions	1
		Rewards for reuse	1
		Software development methodology type	2
	Risk factors	Performance estimation risk process	3
		Requirements instability	2
		Requirements unanalyzability	2
		Requirements uncertainty	1
		Requirement understanding	1
		Risk factors due to task size and team member's experience	1
		Risk perception	1
		Technology uncertainty	1
	Etc.	Absorptive capacity	2
		Code reusability	2
		Competitive advantage of development team	1
		Group coordination	1
		IS project success and performance	11
		Obligation to communicate clearly	1
		Open source software beliefs	1
		Open source software values	1
		Planning analysis	1
		Product/service portfolio differentiation/competitiveness in the market	1
		Project resource availability	4
		Staff skills team and leadership - knowledge & expertise	4
		System outcome	4
		Technology: customized versus canned/generic	1
		Technology evaluation competence	1
		User-IS function relationship	4

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
IT adoption	Affective factors	Affect towards technology use	28
		Attitude towards technology usefulness	7
		Computer alienation	5
		Computer playfulness	4
		Focused immersion	6
		Importance of entertainment	1
		Personal Innovativeness	4
		Result demonstrability	4
		Technology escapism	1
		Technology invasion	1
		Technology loyalty	2
		Technology overload	1
		Technology visibility	2
	Efficacy factors	Computer anxiety	4
		Effort requirement	2
		Knowledge self-efficacy	1
		Lack of adequate skills meaninglessness	1
		Perceived control	4
		Technology-complexity inadequacy	2
		Technology knowledge	3
		Technology self-efficacy	12
	Social/ external factors	Absorptive Capacity	1
		Facilitating conditions	23
		Image	7
		Mimetic pressures	1
		Normative pressure	3
		Readiness of suppliers to do e-business	1
		Social influence	10
	Technology factors	Cognitive challenge of technology	1
		Ease of use	46
		E-shopping versus in-store shopping similarity	1
		Expectations from use of technology	5
		Flexibility, Organization	2
		IT support for contextualization	1
		IT performance	2
		Perceived monetary value	1
		Satisfaction with technology	15
		Social presence of technology	2
		System reliability	3
		Technology availability/accessibility	2
		Technology compatibility with prior experience	2
		Technology diagnosticity	5
		Technology encourages innovation	2
		Flexibility, Individual	4
		Technology functionality perceived	7
		Technology speed	4
		Technology quality	1
		Technology typicality	1

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
IT adoption (continued)	Technology factors (continued)	Trialability	3
		Usefulness to individual	67
		Usefulness to organization	12
		Visual appeal	5
	Use factors	Attitude toward IT	1
		Consensus on appropriation	1
		Faithfulness of appropriation	2
		Habit	1
		Individual use	1
		Intention to use	29
		IT use for specific purposes	7
		Resistance to use	1
		Search scope	2
		Self-reported use	5
		Switching costs	3
		Voluntariness	2
	Etc.	Convenience and confidence	2
		Importance of ease of use	1
		Importance of technology quality	1
		Organizational adoption enabling factors	2
		Technology compatibility with work style & morals/values	10
		Technology confirmation	5
		Time resources/constraints	1
IS function		Adequacy, quality, and amount of support	1
		Complexity evaluation of decision/process	1
		Departmental shared knowledge	1
		Features that determine service quality	2
		Importance of IS department attributes	3
		Importance of IT use in meeting organizational objectives	1
		IS strategic alignment	2
		IT strategy process uniqueness	1
		Mutual influence	1
		Organizational IT capability	7
		Organizational systems integration	2
		Process alignment	2
		Product design maturity	1
		Service quality: Empathy	5
		Service quality: General	3
		Service quality: Reliability	4
		Service quality: Responsiveness	4
		Technology standards	1
		Technology uncertainty	1
Judgement and decision making		Satisfaction with brainstorming outcome	3
		Satisfaction with decision making outcome	8
		Team performance	2

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
Knowledge		Diversity of knowledge	1
		Importance of IS knowledge/skills in business	2
		Importance of IS knowledge in technology	4
		Interorganizational knowledge creation	2
		IS/IT management knowledge of business practices	4
		Knowledge of access to knowledge	1
		Knowledge of team's abilities	1
		Top management IT/IS knowledge	2
Leadership		Importance of leadership skills	2
		Leader effectiveness	1
		Leader/subordinate relationship	1
		Organizational skills	1
		Personal responsibility for leadership	3
		Power	1
		Presence of leadership skills	1
Learning		Attitude toward knowledge sharing	1
		Course enjoyment	1
		Group work contributes to learning	2
		Learning frequency	1
		Learning orientation	1
		Motivation to learn	2
		Observational learning process	3
		Organizational learning	1
		Perceived learning	6
		Relative advantage	1
		Self-efficacy for doing school work	1
		Self-regulation during learning	1
		Skill development	1
		Subjective norm toward knowledge sharing	1
Organizational	Financials	Business performance	3
		Task non-contractability quality	2
	Money	Market orientation	1
	Organization-level characteristics	Business-process specificity: Human capital	2
		Business-process specificity: Intellectual capital	3
		Formalization	2
	People	Impact of system on organization	1
		Interactions between CIO and TMT	2
		IT hiring and retainment practices	1
	Perceptions of organization	Ambiguity	1
		Benefits uncertainty	1
		Fairness of labor division	1
		Information need fulfillment	1
		Norm of cooperation	1
		Organizational commitment	4
		Organizational uncertainty: Market turbulence	4
		Organizational uncertainty: Technological uncertainty	1
		Organizational uncertainty: Production volume	1
		Perceived competitive advantage	4

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
Organizational (continued)	Perceptions of organization (continued)	Perceived strategic risks	1
		Perceived supportive affect within work groups	1
		Work environment satisfaction	1
Privacy		Benefit overriding privacy concerns	2
		Concern over unauthorized access	3
		Concern over unauthorized secondary use	2
		Errors in private data/info	3
		General concerns about information privacy	16
		Importance of transparency info use policies	2
		Perceived protection by companies	2
		Personal control of information	2
		Risk of loss	2
		Technology security risk	2
		User awareness of security policies	3
Purchase		Buyer's intention to purchase	2
		Consumer's opinions of traveling to make a purchase	1
		Importance of shopping convenience	1
		Internet customer relation	2
		Monetary resources	1
		Need for uniqueness from purchasing	1
		Perceived product choice on the internet	1
		Product value	2
		Purchasing skills	1
		Satisfaction with purchasing experience	6
		Seller's past performances	3
		Societal benefit from complaining	1
Strategy		Business-process strategic criticality	2
		Locus of authority for strategic planning	1
		Strategic aggressiveness	1
		Strategic defensiveness	1
		Strategic intent	1
		Strategic investment rationale	1
		Strategy comprehensiveness	1
		Strategy proactiveness	1
		Strategy: Risk aversion	1
Task/job	Employment	Job turnover intention	3
	Job specifics	Fairness	2
		Improved job performance due to autonomy	2
		Job overload	9
	Outcomes/Performances/ Rewards	Benefit of efforts	1
		Determination of rewards	1
	Satisfaction	Individual rewards present	1
		Job satisfaction	2
		Rewards	2
	Task characteristics	Business process modularity	1
		Job autonomy	3
		Role ambiguity	2
		Role/task conflicts	3

Table B1. The Gold Standard Taxonomy (Continued)

Hierarchy	Sub-hierarchy	Category	# Constructs
Task/job (continued)	Task characteristics (continued)	Task difficulty	1
		Task interdependence	1
		Task uncertainty	1
		Task variety	4
		Uncertainty with information	1
	Etc.	Emotion toward customer	2
		Work-family conflict	3
Trust		Credibility Trust in ability and benevolence	9
		Individual-level trust: Other's ability	1
		Individual-level trust: Other's ability and trustworthiness	3
		Individual-level trust: Other's Benevolence	1
		Individual-level trust: Structural assurances	5
		Individual-level trust: Trust propensity	9
		Individual-level trust: Trustworthiness	19
		Organization-level trust: Other's ability	2
		Organization-level trust: Other's benevolence	2
		Organization-level trust: Other's integrity	2
		Organization-level trust: Reputation	2
		Trust in others' ability	8

Appendix C

Item to Construct Solutions

This appendix begins by examining approaches to the transformation of similarity scores from the item level to the construct level. It then discusses how such transformations may affect the analysis of formative versus reflective items.

Generalized Functions for Transforming Similarity Scores from the Item Level to the Construct Level

Regardless of the designs used, construct relationships can be predicted by comparing two items, where each construct item is treated as natural text and a similarity score is produced to indicate their degree of similarity. Item similarities are computed for all inter-construct item pairs, and the item scores are transformed to indicate a construct relationship. Two functions are formulated to subsume item similarities into the construct relationship. While future work should examine all possibilities, the first measure (*sec*) tested for this article used the second highest score among all inter-construct item relationships. This function is based on the notion that the two constructs' highest inter-item relationship can be misleading, while the use of the second highest relationship score results in a compromise between the rate of false positives and false negatives. The second function used to represent the construct similarity (*avg*) takes the average of the similarity of the most similar item pair scores. This function tends to compensate for item pairs that are constituted of both highly similar and very dissimilar constituent parts. Although we examined both measures, we anticipated that *avg* could better represent the construct relationship without the skewedness demonstrated by *sec*.

An analysis evaluating *sec* versus *avg* for LSA showed that both generalized functions were above the diagonal line (> 0.5), indicating that both functions *sec* (AUC = 0.759) and *avg* (AUC = 0.767) used after LSA performed better than chance. CROC tests, with $\alpha < 0.01$ against both generalized LSA functions, showed that both functions performed significantly better than chance and that the *avg* function significantly outperformed the *sec* function. This pattern was confirmed for tests on all algorithms. We show further evaluations only with the *avg* function results.

Reflective Versus Formative Constructs

The majority of constructs that we included in this study were reflective constructs such as *ease of use*, where the items are assumed to reflect an underlying latent unobservable phenomenon (Petter et al. 2007). Whether examining reflective constructs, composites of multiple measures (Petter et al. 2007), or formative constructs, different construct types may be measured using identical items. In our research, we came across several examples of this, including the measurement of *privacy concerns* (defined as "subjects' concerns with their privacy over the Internet") (Hui et al. 2007) versus *improper access* ("consumers' opinions about improper access in organizational practices") (Malhotra et al. 2004). Both of these constructs are measured by multiple items, including the shared item, "Computer databases that contain personal information should be protected from unauthorized access—no matter how much it costs."

Figure C1 illustrates how CID operates to overcome some of these problems. In the figure, reflective constructs A and B (for example, two different *ease of use* constructs) have three items each. After evaluating all nine potential inter-item relationships for the two constructs, CID detects and links the three highest-similarity item pairs, starting with the most similar pairs. CID then uses the average similarity score as an evaluator of the construct relationship. CID conducts the same evaluations between all four constructs, resulting in the detection of high-similarity item pairs between two of the three items for formative constructs X and Y (for example, user information satisfaction [UIS], which often contains an *ease of use* item and otherwise taps into a number of different phenomena). Of the formative constructs in the example, only construct X references an ease-of-use item (X_1).

In the example, X_1 is highly similar to all items for reflective constructs A and B, but it is allowed to establish a relationship only to its highest-similarity item within each alternative construct. For example, for both reflective and formative constructs the most similar items are connected. What this means is that even when a formative construct contains items that are quite dissimilar from other items within the construct, if another formative construct, contains identical or similar items (also dissimilar within the construct), the relationship between the two formative constructs will still be high, because only the most similar items are connected between the two. If the two formative constructs focus on different phenomenon, while any linked item pair will be the *most* similar of the pairs, it will still be very dissimilar, leading to a low inter-construct relationship. Per Figure C1, the relationship between items in X and B may be X_1—$B_1 = 1$, X_2—$B_3 = 0$, X_3—$B_2 = 0$). If each high-

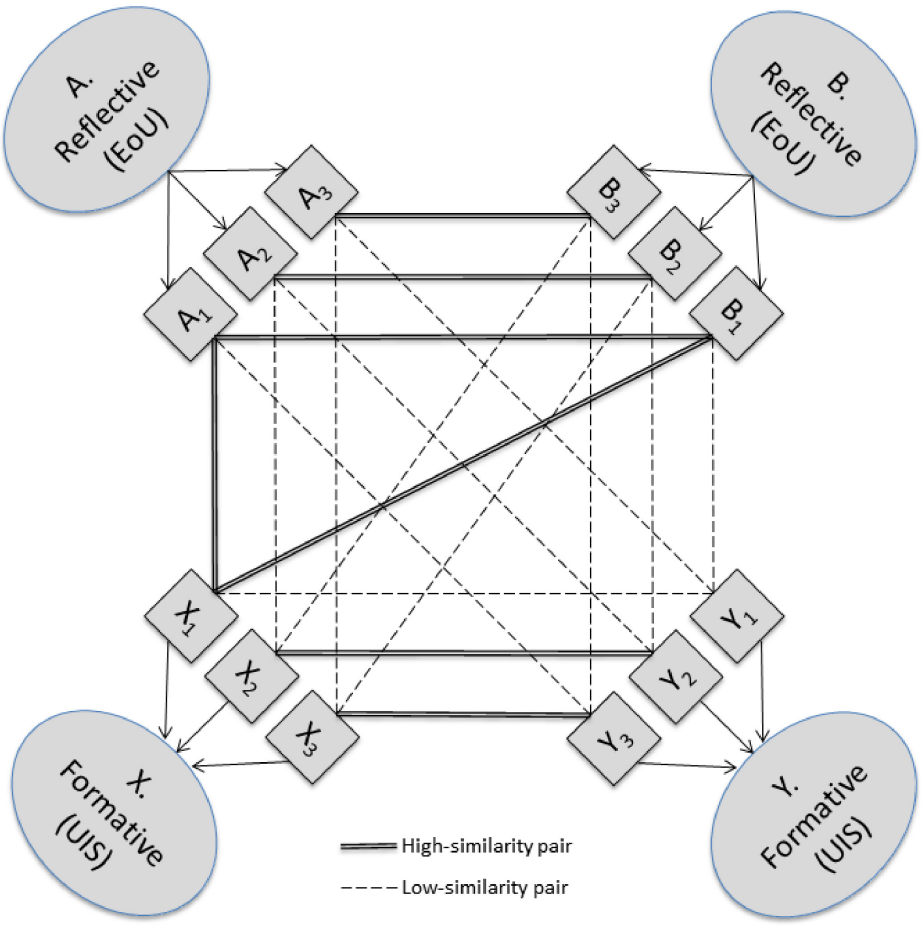


Figure C1. CID on Reflective Versus Formative Constructs

similarity pair (double lines) is scored as 1.0 and each low-similarity pair (dotted lines) is scored 0, the construct–construct scores become A—B = 1, X—Y = .66, X—A = .33, X—B = .33, Y—A = 0, and Y—B = 0. In short, the process is relatively invariant in terms of reflective and formative constructs. That is, it should work equally well for either construct type while exhibiting low-to-moderate relationships between reflective and formative constructs.

Appendix D

Evaluation of Automatic Cutoffs

To evaluate the potential impact of using the automatic F_1 cutoffs when evaluating CID_1, we conducted a separate assessment, in which a random draw resulted in 20 constructs that were used as starting points to find all constructs in the same category. We used CID_1 in the same manner as the other tests documented in this article, whereas two approaches were used to query EBSCO for each of the 20 construct categories. In the first approach, the words from the constructs' measurement items were weighted using log-entropy and the weights combined across all items in the construct; the top three words were then used to create queries. In the second approach, the name of the construct was used as a search query in up to two combinations for longer construct names. These procedures produced up to four queries for each EBSCO

construct search task, of which the best performing query, as measured by maximum F_1-score, was elected to represent expert queries. The randomly selected tasks were expected to return between 1 and 56 constructs (average = 18.1 and median = 6.5, suggesting a few large tasks and several smaller tasks). CID_1 outperformed EBSCO on every assessment measure: average precision (.68 versus .46), average recall (.58 versus .39), average F_1 (.60 versus .32), average constructs found (7.50 versus 3.55), and percent of constructs found (52.82 versus 28.85), thereby supporting the original findings.

Appendix E

Supervised Machine Learning for CID

To examine the potential of supervised machine learning for CID, the six similarity designs were used to create input features for supervised machine learning using the gold standard as the target. Using the five-fold sampling process on construct pairs leads to problems because any randomly selected 80% of pairs (36 pairs) would still leave the network almost intact and result in a test set that would incorrectly represent the algorithm's predictive abilities. Therefore, a fully categorized sample of constructs was not an option. For a relatively simple test, we started with the gold standard of 1,004 constructs and randomly split the sample into five sets of 504 constructs for training (126,756 pairs) and 504 constructs for testing (again with 126,756 pairs). For each pair, the six similarity measures were used as features in the training and evaluation, and the synonymy status of each pair as the target variable.

Table E1. Evaluative Measures for Supervised Learning

J48	Precision	Recall	F_1	Correspondent Constructs Found	Total Constructs Returned
Sample 1	0.715	0.182	0.29	246	344
Sample 2	0.594	0.199	0.298	259	436
Sample 3	0.781	0.121	0.21	210	269
Sample 4	0.605	0.159	0.252	187	309
Sample 5	0.603	0.135	0.22	173	287
Average	**0.659**	**0.159**	**0.254**	**215**	**329**
Random Forest					
Sample 1	0.783	0.196	0.313	314	401
Sample 2	0.638	0.169	0.267	220	345
Sample 3	0.696	0.127	0.215	220	316
Sample 4	0.596	0.145	0.233	170	285
Sample 5	0.651	0.148	0.241	190	292
Average	**0.672**	**0.157**	**0.253**	**222.8**	**327.8**
Naïve Bayes					
Sample 1	0.146	0.416	0.216	636	4348
Sample 2	0.12	0.441	0.188	575	4799
Sample 3	0.147	0.407	0.216	705	4799
Sample 4	0.098	0.417	0.159	490	4982
Sample 5	0.111	0.45	0.178	579	5234
Average	**0.124**	**0.426**	**0.191**	**597**	**4832.4**
CID_1					
Sample 1	0.322	0.259	0.287	397	1232
Sample 2	0.411	0.258	0.317	336	818
Sample 3	0.4111	0.24	0.303	416	1011
Sample 4	0.373	0.239	0.291	281	754
Sample 5	0.409	0.255	0.314	328	803
Average	**0.385**	**0.250**	**0.302**	**351.6**	**923.6**

For each training set, a predictive model was created using three Weka version 3.7.9 algorithms with default parameters: *Naïve Bayes, J48,* and *Random Forest* (Bouckaert et al. 2013; Hall et al. 2009).[2] The test datasets were used to evaluate the success of the three algorithms against the success of the CID_1 design. As may be seen from Table E1, while CID_1 outperforms the other algorithms—especially Naïve Bayes—substantially on average F_1-scores, the precision of both J48 and Random Forest is quite impressive. Unfortunately, the recall scores for both algorithms are low. Nevertheless, the results suggest that work on supervised machine learning holds great potential for future improvements.

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